Rockchip RKADK Development Guide

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Preface

Overview

This document is going to introduce development reference of Rockchip Rkadk component.

Product Version

Chipset	Kernel Version
RV1126/RV1109	Linux 4.19
RV1106/RV1103	Linux 5.10
RV1103B	Linux 5.10
RK3576	Linux 6.10
RV1126B	Linux 6.10

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Change Description
V1.0.0	CTF	2021-05-02	Initial version
V1.1.0	CTF	2021-08-23	1. Add API introduction: (1) RKADK_RECORD_GetAencChn (2) RKADK_GetThmInMp4 (3) RKADK_PHOTO_GetThmInJpg (4) RKADK_PARAM_GetVencChnId 2. New module introduction: (1) Local preview module `Display` (2) Live broadcast module `Live`: includes RTSP and RTMP sub-modules
V1.2.0	CTF	2021-09-26	1. Add thumbnail extension API: (1) RKADK_GetThmInMp4Ex (2) RKADK_ThmBufFree (3) RKADK_PHOTO_GetThmInJpgEx (4) RKADK_PHOTO_ThumbBufFree 2. Support multi sensor 3. Support setting INI file path 4. Add VI ISP module 5. Support setting VENC GOP
V1.3.0	GZC	2021-11-30	New storage module
V1.3.1	CTF	2021-12-20	1. Add API introduction: (1) RKADK_PHOTO_GetData (2) RKADK_PHOTO_FreeData (3) RKADK_PLAYER_GetDuration 2. Add INI configuration notes
V2.0.0	CTF	2023-01-29	1. Adapt to general Linux SDK 2. Add API introduction: (1) RKADK_OSD_Init (2) RKADK_OSD_Deinit (3) RKADK_OSD_UpdateOsdSize (4) RKADK_OSD_AttachToStream (5) RKADK_OSD_DettachFromStream (6) RKADK_OSD_UpdateDisplayAttr
V2.1.0	CTF	2023-04-27	Compatible with RV1109/RV1126 platforms

Version	Author	Date	Change Description
V2.2.0	CTF	2023-11-07	1. Added UI overlay module 2. Added JPEG Slice function 3. Added the following functions to Player: (1) Supported third-party demuxer library (2) Supports RTSP network stream playback (3) Supports screen snapshot function (4) Compatible with RV1106/RV1103, RK3308, RK3506 platforms 4. Added Record File Cache description 5. Add Record/Photo rotation, flip, mirror API description 6. Update ini description 7. Update test cases
V2.2.1	CTF	2024-01-04	1. Added Post AI ISP function
V2.2.2	CTF	2024-04-23	Added AOV(Always On Video) time-lapse recording function
V2.2.3	CTF	2024-09-11	1. Added Picture-In-Picture (PiP) recording function 2. Add API introduction: (1) RKADK_RECORD_Single_Start (2) RKADK_RECORD_Single_Stop (3) RKADK_RECORD_SetPipAttr (4) RKADK_RECORD_FileCacheSetMode (5) RKADK_PLAYER_GetSendFrameNum (6) RKADK_PLAYER_SetVdecWaterline (7) RKADK_PLAYER_SetAoVolume
V2.3.0	CTF	2025-04-07	1.Compatible with RV1103B, RK3576, RV1126B platforms 2. Add API introduction: (1) RKADK_STREAM_VideoReset (2) RKADK_RTSP_VideoReset (3) RKADK_RTMP_VideoReset

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1. Overview

Rkadk provides basic and universal components such as video recording, photo capture, playback, preview, etc., simplifying the difficulty of application development and supporting the rapid development of audio and video recording-related software.

This component only supports the implementation of single-process functionality. It does not support multiple processes used simultaneously unless noted otherwise.

1.1 Version Introduction

- Version 1.x.x: Further encapsulation based on rkmedia and rockit, corresponding to the master repository.
- Version 2.x.x: Further encapsulation based on rockit, corresponding to the develop repository.

2. Video Recording

2.1 Overview

Provides basic video recording functionality, offering the following features to product:

- Creation and destruction of recording tasks
- Start and stop of recording tasks
- Manual slice of recording files
- Time-lapse recording
- Pre-recording
- Picture-In-Picture (PiP) recording

Recording tasks obtain video and audio information through the parameter module, start and stop VENC (video encoding), start and stop AENC (audio encoding), and call the encapsulation module to create recording files and write frames to the files.

Each recording task corresponds to one or more recording files. Each file must correspond to a video encoding channel, and if audio recording is needed, an audio encoding channel must be added.

Multiple recording files under the same recording task have the same recording type, the same slicing conditions, and in time-lapse recording mode, they also have the same time-lapse interval. Different recording times can be configured.

Post AI ISP: RV1106/RV1126/RV1109 chip supports noise reduction in dim light environments and intelligent enhancement processing of VI output images through AI ISP, which can still present a clearer picture without dragging shadows, low noise, and even in dim light environments.

2.2 API Reference

2.2.1 RKADK_RECORD_Create

Description

Create a video recording task.

Syntax

RKADK_S32 RKADK_RECORD_Create(RKADK_RECORD_ATTR_S *pstRecAttr, RKADK_MW_PTR *ppRecorder);

Parameters

Parameter name	Description	Input/output
pstRecAttr	Recording task attributes	Input
ppRecorder	Created recording task pointer	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

Notice

- Each recording task supports a maximum of 2 recording files simultaneously.
- Each recording file must have at least 1 video stream and supports encapsulating 1 video stream and 1 audio stream simultaneously at most.
- It does not support the creation of the same task repeatedly.
- After creating a recording task, RKADK_RECORD_Start must be called to start recording.
- To enable the Picture-in-Picture feature, it must be enabled through stPipAttr when RKADK_RECORD_Create, which will then create the Picture-in-Picture pathway.

Example

rkadk_record_test.

See Also

RKADK_RECORD_Destroy.

2.2.2 RKADK_RECORD_Destroy

Description

Destroy video tasks.

Syntax

RKADK_S32 RKADK_RECORD_Destroy(RKADK_MW_PTR pRecorder);

Parameters

Parameter name	Description	Input/output
pRecorder	Pointer to recording task	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

Notice

- The RKADK_RECORD_Destroy interface can only be used after creating a recording task.
- Only created recording tasks can be destroyed, and repeated destruction of the same recording task is not supported.

Example

rkadk_record_test.

See Also

RKADK_RECORD_Create

2.2.3 RKADK_RECORD_Start

Description

Start recording task.

Syntax

RKADK_S32 RKADK_RECORD_Start(RKADK_MW_PTR pRecorder);

Parameters

Parameter name	Description	Input/output
pRecorder	Pointer to recording task	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

Notice

- The RKADK_RECORD_Start interface can only be used after creating a recording task.
- It is possible to restart a recording task after stopping it.

Example

rkadk_record_test.

See Also

RKADK_RECORD_Stop

2.2.4 RKADK_RECORD_Stop

Description

Stop recording task.

Syntax

RKADK_S32 RKADK_RECORD_Stop(RKADK_MW_PTR pRecorder);

Parameters

Parameter name	Description	Input/output
pRecorder	Pointer to recording task	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

Notice

- The RKADK_RECORD_Stop interface can only be used after creating a recording task.
- Repeatedly stopping the same recording task is not supported.

Example

rkadk_record_test.

See Also

RKADK_RECORD_Start

2.2.5 RKADK_RECORD_Single_Start

Description

Start the recording task for a specified stream.

Syntax

RKADK_S32 RKADK_RECORD_Single_Start(RKADK_MW_PTR pRecorder, RKADK_STREAM_TYPE_E enStrmType);

Parameters

Parameter Name	Description	Input/Output
pRecorder	Pointer to recording task	Input
enStrmType	Stream type	Input

Return Value

Return Value	Description
0	Success
Other value	Failure

Requirements

Header file: rkadk_record.h

Library file: librkadk.so

Notes

- The RKADK_RECORD_Single_Start interface can only be used after creating a recording task.
- Used in conjunction with RKADK_RECORD_Single_Stop.
- Supports restarting the recording task after stopping.

Example

rkadk_record_test.

Related Topics

RKADK_RECORD_Single_Stop.

2.2.6 RKADK_RECORD_Single_Stop

Description

Stop the recording task for a specified stream.

Syntax

RKADK_S32 RKADK_RECORD_Single_Stop(RKADK_MW_PTR pRecorder, RKADK_STREAM_TYPE_EenStrmType);

Parameters

Parameter Name	Description	Input/Output
pRecorder	Pointer to recording task	Input
enStrmType	Stream type	Input

Return Values

Return Value	Description
0	Success
Other value	Failure

Requirements

Header file: rkadk_record.h

Library file: librkadk.so

Notes

- The RKADK_RECORD_Stop interface can only be used after creating a recording task.
- Use in conjunction with RKADK_RECORD_Single_Start.
- Does not support stopping the same recording task repeatedly.

Example

rkadk_record_test.

Related Topics

RKADK_RECORD_Single_Start

2.2.7 RKADK_RECORD_Reset

Description

Reconfigure the recording task.

Syntax

RKADK_S32 RKADK_RECORD_Reset(RKADK_MW_PTR *ppRecorder);

parameters

Parameter name	Description	Input/output
ppRecorder	Pointer to recording task	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

Notice

- The RKADK_RECORD_Reset interface can only be used after creating a recording task.
- When switching resolution, frame rate, bit rate, encoding type, or recording type, reset the parameters of the Record module in the ini file by using the RKADK_PARAM_SetCamParam API.
- RV1126/RV1109 and RK3576 does not support dynamic switching of resolution and encoding type. You need to RKADK_RECORD_Destroy first, then configure the new resolution or encoding type, and finally RKADK_RECORD_Create again.
- If Record and other modules share the VI channel, both the resolution of Record and multiplexing modules need to be switched simultaneously to avoid VI and VENC resolution mismatch.

Example

rkadk_record_test.

See Also

None

2.2.8 RKADK_RECORD_ManualSplit

Description

Manually split video files.

Syntax

RKADK_S32 RKADK_RECORD_ManualSplit(RKADK_MW_PTR pRecorder,
RKADK_REC_MANUAL_SPLIT_ATTR_S *pstSplitAttr);

Parameters

Parameter name	Description	Input/output
pRecorder	Pointer to recording task	Input
pstSplitAttr	Manual split attribute parameters	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

Notice

- The RKADK_RECORD_ManualSplit interface can only be used after creating a recording task.
- Supports repeated manual splitting of video files when the manual splitting of video files has not ended.

Example

rkadk_record_test.

See Also

No

2.2.9 RKADK_RECORD_GetAencChn

Description

Get the video AENC channel ID.

Syntax

RKADK_S32 RKADK_RECORD_GetAencChn();

Return value

AENC channel ID used for recording.

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

2.2.10 RKADK_GetThmInMp4

Description

Get thumbnail data from MP4 files.

Syntax

```
RKADK_S32 RKADK_GetThmInMp4(RKADK_U32 u32CamId, RKADK_CHAR *pszFileName,
RKADK_U8 *pu8Buf,
                            RKADK_U32 *pu32Size);
```

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
pszFileName	MP4 file path	Input
pu8Buf	Input: thumbnail data storage pointer, output: actual thumbnail data	Input/Output
pu32Size	Input: pu8Buf length, output: actual thumbnail data length	Input/Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_thumb.h

Library file: librkadk.so

Example

rkadk_thumb_test

See Also

None

2.2.11 RKADK_GetThmInMp4Ex

Description

MP4 obtains thumbnail extension interface, obtains thumbnail data from MP4 files, and supports specifying the type and resolution of output thumbnails. It must be used together with RKADK_ThmBufFree.

Syntax

RKADK_S32 RKADK_GetThmInMp4Ex(RKADK_U32 u32CamId, RKADK_CHAR *pszFileName, RKADK_THUMB_ATTR_S *pstThumbAttr);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
pszFileName	MP4 file path	Input
pstThumbAttr	Thumbnail attribute structure pointer	Input/output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_thumb.h

Library file: librkadk.so

Example

rkadk_thumb_test

See Also

RKADK_ThmBufFree

2.2.12 RKADK_ThmBufFree

Description

Release the memory requested by RKADK_GetThmInMp4Ex. It must be used together with RKADK_GetThmInMp4Ex.

Syntax

RKADK_S32 RKADK_ThmBufFree(RKADK_THUMB_ATTR_S *pstThumbAttr);

Parameters

Parameter name	Description	Input/output
pstThumbAttr	Thumbnail attribute structure pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_thumb.h

Library file: librkadk.so

Example

rkadk_thumb_test

See Also

RKADK_GetThmInMp4Ex

2.2.13 RKADK_RECORD_SetRotation

Description

Set Record rotation.

Syntax

RKADK_S32 RKADK_RECORD_SetRotation(RKADK_MW_PTR pRecorder, ROTATION_E enRotation, RKADK_STREAM_TYPE_E enStreamType);

Parameters

Parameter name	Description	Input/output
pRecorder	Pointer to recording task	Input
enRotation	Rotation angle	Input
enStreamType	Stream type	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

Example

rkadk_record_test

2.2.14 RKADK_RECORD_ToggleMirror

Description

Set Record Mirror.

Syntax

RKADK_S32 RKADK_RECORD_ToggleMirror(RKADK_MW_PTR pRecorder, RKADK_STREAM_TYPE_E enStrmType, bool mirror);

Parameters

Parameter name	Description	Input/output
pRecorder	Pointer to recording task	Input
enStreamType	Stream type	Input
mirror	Whether to enable mirror	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

2.2.15 RKADK_RECORD_ToggleFlip

Description

Set Record Flip.

Syntax

RKADK_S32 RKADK_RECORD_ToggleFlip(RKADK_MW_PTR pRecorder, RKADK_STREAM_TYPE_E enStrmType, bool flip);

Parameters

Parameter name	Description	Input/output
pRecorder	Pointer to recording task	Input
enStreamType	Stream type	Input
flip	Whether to enable flip	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

2.2.16 RKADK_RECORD_FileCacheInit

Description

File Cache initialization is aimed at making file writing smoother. Enabling File Cache means that files are written to the storage device through DirectIO, and the kernel's cache mechanism is not effective.

Syntax

RKADK_S32 RKADK_RECORD_FileCacheInit(FILE_CACHE_ARG *pstFileCacheAttr)

Parameters

Parameter name	Description	Input/output
pstFileCacheAttr	File Cache attribute pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

Notice

- File Cache is not enabled by default.
- To enable File Cache, the file_cache_env environment variable must be set

```
export file_cache_env=1
```

- If File Cache is enabled, this interface must be called before RKADK_RECORD_Create, and only needs to be called once, otherwise there is no need to call it.
- Enabling File Cache will need additional memory, and the memory size is determined by u32TotalCache.
- This interface must be used in pair with RKADK_RECORD_FileCacheDeInit

Example

rkadk_record_test

2.2.17 RKADK_RECORD_FileCacheDeInit

Description

De-initialization File Cache

Syntax

RKADK_S32 RKADK_RECORD_FileCacheDeInit();

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_record.h

Library file: librkadk.so

Notice

- It must be used together with RKADK_RECORD_FileCacheInit.
- If File Cache is enabled, this interface needs to be called after RKADK_RECORD_Destroy, otherwise there is no need to call it.

rkadk_record_test

2.2.18 RKADK_RECORD_FileCacheSetMode

Description

Set the recording type for File Cache.

Syntax

void RKADK_RECORD_FileCacheSetMode(RKADK_REC_TYPE_E enRecType);

Parameters

Parameter Name	Description	Input/Output
enRecType	Recording type	Input

Requirements

Header file: rkadk_record.h

Library file: librkadk.so

Notes

- After enabling File Cache, when switching recording types, this API must be called to set a new recording type for File Cache.

rkadk_aov_record_test

2.2.19 RKADK_MEDIA_EnablePostIsp

Description

Enable post AI ISP, in addition to the interface, you can also enable post AI ISP by configuring post_aiisp in sensor ini.

Syntax

RKADK_S32 RKADK_MEDIA_EnablePostIsp(RKADK_U32 u32CamId, RKADK_STREAM_TYPE_E enStrmType, RKADK_POST_ISP_ATTR_S *pstPostIspAttr);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
enStrmType	Stream type	Input
pstPostIspAttr	Post AI ISP attribute pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_media_comm.h

Library file: librkadk.so

2.2.20 RKADK_MEDIA_SetPostIspAttr

Description

Dynamically set post AI ISP attributes.

Syntax

RKADK_S32 RKADK_MEDIA_SetPostIspAttr(RKADK_U32 u32CamId, RKADK_STREAM_TYPE_E enStrmType, bool bEnable, RKADK_POST_ISP_ATTR_S *pstPostIspAttr);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
enStrmType	Stream type	Input
bEnable	Post AI ISP function switch, dynamic attribute	Input
pstPostIspAttr	Post AI ISP attribute pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_media_comm.h

Library file: librkadk.so

2.2.21 RKADK_RECORD_SetPipAttr

Description

Dynamically set picture-in-picture (PiP) attributes.

Syntax

RKADK_S32 RKADK_RECORD_SetPipAttr(RKADK_MW_PTR pRecorder, RKADK_PIP_ATTR_S *pstPipAttr);

Parameters

Parameter Name	Description	Input/Output
pRecorder	Pointer to recording task	Input
pstPipAttr	Picture-in-picture attributes	Input

Return Value

Return Value	Description
0	Success
Other value	Failure

Requirements

Header file: rkadk_record.h

Library file: librkadk.so

Note

- To enable the Picture-in-Picture feature, it must be enabled through stPipAttr when RKADK_RECORD_Create, which will then create the Picture-in-Picture pathway.

2.3 Type of Data

The video recording module mainly provides the following data types:

RKADK_MW_PTR: Pointer to recording task

RECORD_FILE_NUM_MAX: The maximum number of files recorded simultaneously by a single recording task

RKADK_MUXER_EVENT_E: Recording event enumeration type

RKADK_MUXER_FILE_EVENT_INFO_S: File-related event information structure

RKADK_MUXER_EVENT_INFO_S: Recording event information structure

RKADK_REC_EVENT_CALLBACK_FN: Event callback function pointer

RKADK_REC_TYPE_E: Recording type enumeration

RKADK_REC_REQUEST_FILE_NAMES_FN: Request video file name function pointer

RKADK_RECORD_ATTR_S: Recording task attribute structure

RKADK_MUXER_MANUAL_SPLIT_TYPE_E: Manually split enumeration types

RKADK_REC_MANUAL_SPLIT_ATTR_S: Manually split attribute structure

FILE_CACHE_ARG: File Cache attribute structure

RKADK_POST_ISP_ATTR_S: Post AI ISP attribute structure

RKADK_MOUNT_SDCARD_FN: SD card mounting function pointer

RKADK_AOV_ATTR_S: AOV attribute structure

RKADK_PIP_ATTR_S: Picture-in-Picture attribute structure

2.3.1 Public Data Types

Description

Definition of basic data types.

Definition

```
typedef unsigned char RKADK_U8;
typedef unsigned short RKADK_U16;
typedef unsigned int RKADK_U32;

typedef signed char RKADK_S8;
typedef short RKADK_S16;
typedef int RKADK_S32;

typedef unsigned long RKADK_UL;
typedef signed long RKADK_SL;

typedef float RKADK_FLOAT;
typedef double RKADK_DOUBLE;

#ifndef _M_IX86
typedef unsigned long long RKADK_U64;
typedef long long RKADK_S64;
#else
typedef unsigned __int64 RKADK_U64;
typedef __int64 RKADK_S64;
#endif

typedef char RKADK_CHAR;
#define RKADK_VOID void

typedef unsigned int RKADK_HANDLE;

typedef RKADK_VOID *RKADK_MW_PTR;

typedef char (*ARRAY_FILE_NAME)[RKADK_MAX_FILE_PATH_LEN];

typedef enum {
  RKADK_FALSE = 0,
  RKADK_TRUE = 1,
} RKADK_BOOL;

#ifndef NULL
#define NULL 0L
#endif

#define RKADK_NULL 0L
#define RKADK_SUCCESS 0
#define RKADK_FAILURE (-1)
```

2.3.2 RKADK_MW_PTR

Description

Define task pointer

Definition

```
typedef RKADK_VOID *RKADK_MW_PTR;
```

2.3.3 RKADK_MAX_SENSOR_CNT

Description

Define the maximum number of sensors supported, which can be adjusted according to the actual situation

Definition

```
#define RKADK_MAX_SENSOR_CNT 3
```

2.3.4 RECORD_FILE_NUM_MAX

Description

Define the maximum number of files recorded simultaneously by a single recording task.

Definition

```
#define RECORD_FILE_NUM_MAX 2
```

2.3.5 RKADK_MUXER_EVENT_E

Description

Define the recording event enumeration type.

Definition

```
typedef enum rkMUXER_EVENT_E {
  RKADK_MUXER_EVENT_STREAM_START = 0,
  RKADK_MUXER_EVENT_STREAM_STOP,
  RKADK_MUXER_EVENT_FILE_BEGIN,
  RKADK_MUXER_EVENT_FILE_END,
  RKADK_MUXER_EVENT_MANUAL_SPLIT_END,
  RKADK_MUXER_EVENT_ERR_CREATE_FILE_FAIL,
  RKADK_MUXER_EVENT_ERR_WRITE_FILE_FAIL,
  RKADK_MUXER_EVENT_FILE_WRITING_SLOW,
  RKADK_MUXER_EVENT_ERR_CARD_NONEXIST,
  RKADK_MUXER_EVENT_BUTT
} RKADK_MUXER_EVENT_E;
```

Members

Member Name	Description
RKADK_MUXER_EVENT_STREAM_START	Start recording
RKADK_MUXER_EVENT_STREAM_STOP	Stop recording
RKADK_MUXER_EVENT_FILE_BEGIN	Start recording a new file
RKADK_MUXER_EVENT_FILE_END	End of file recording
RKADK_MUXER_EVENT_MANUAL_SPLIT_END	End of manual split file recording
RKADK_MUXER_EVENT_ERR_GET_FILENAME	Failed to get file name
RKADK_MUXER_EVENT_ERR_CREATE_FILE_FAIL	Reserved
RKADK_MUXER_EVENT_ERR_WRITE_FILE_FAIL	Failed to write file
RKADK_MUXER_EVENT_FILE_WRITING_SLOW	Writing files is slow
RKADK_MUXER_EVENT_ERR_CARD_NONEXIST	SD card does not exist

Related data types and interfaces

RKADK_MUXER_EVENT_INFO_S

2.3.6 RKADK_MUXER_FILE_EVENT_INFO_S

Description

Define file-related event information structure.

Definition

```
typedef struct {
  RK_CHAR asFileName[RKADK_MUXER_FILE_NAME_LEN];
  RK_U32 u32Duration; // ms
} RKADK_MUXER_FILE_EVENT_INFO_S;
```

Members

Member Name	Description
asFileName	File name
u32Duration	Actual recorded file duration

Related data types and interfaces

RKADK_MUXER_EVENT_INFO_S

2.3.7 RKADK_MUXER_EVENT_INFO_S

Description

Define the recording event information structure.

Definition

```
typedef struct {
  RKADK_MUXER_EVENT_E enEvent;
  union {
    RKADK_MUXER_FILE_EVENT_INFO_S stFileInfo;
    RKADK_MUXER_ERROR_EVENT_INFO_S stErrorInfo;
  } unEventInfo;
} RKADK_MUXER_EVENT_INFO_S;
```

Members

Member Name	Description
enEvent	Recording event type
stFileInfo	File event information
stErrorInfo	Error event information (Reserved)

Related data types and interfaces

RKADK_MUXER_EVENT_E

RKADK_MUXER_FILE_EVENT_INFO_S

2.3.8 RKADK_REC_EVENT_CALLBACK_FN

Description

Define the video event callback function pointer.

Definition

```
typedef RKADK_MUXER_EVENT_CALLBACK_FN RKADK_REC_EVENT_CALLBACK_FN;

typedef RKADK_VOID (*RKADK_MUXER_EVENT_CALLBACK_FN)(RKADK_MW_PTR pRecorder, const
RKADK_MUXER_EVENT_INFO_S *pstEventInfo);
```

Related data types and interfaces

RKADK_MW_PTR

RKADK_MUXER_EVENT_INFO_S

2.3.9 RKADK_REC_TYPE_E

Description

Define the enumeration of recording types.

Definition

```
typedef enum {
  RKADK_REC_TYPE_NORMAL = 0, /* normal record */
  RKADK_REC_TYPE_LAPSE,      /* time lapse record */
  RKADK_REC_TYPE_AOV_LAPSE,  /* low power time lapse record */
  RKADK_REC_TYPE_BUTT
} RKADK_REC_TYPE_E;
```

Members

Member Name	Description
RKADK_REC_TYPE_NORMAL	Normal recording
RKADK_REC_TYPE_LAPSE	Time-lapse video
RKADK_REC_TYPE_AOV_LAPSE	Low power time-lapse record

Related data types and interfaces

None

Note

- Only RV1126 and RV1106 platforms support AOV time-lapse record

2.3.10 RKADK_REC_REQUEST_FILE_NAMES_FN

Description

Define the callback function pointer to request the video file name.

Definition

```
typedef RKADK_S32 (*RKADK_REC_REQUEST_FILE_NAMES_FN)(RKADK_MW_PTR pRecorder,
RKADK_U32 u32FileCnt, RKADK_CHAR(*paszFilename)[RKADK_MAX_FILE_PATH_LEN]);
```

Members

Member Name	Description
pRecorder	Pointer to recording task
u32FileCnt	Number of requested file names
paszFilename	Store file name buffer

Related data types and interfaces

RKADK_MW_PTR

RKADK_RECORD_ATTR_S

2.3.11 RKADK_MOUNT_SDCARD_FN

Description

Define the SD card mounting function pointer.

Definition

```
typedef int (*RKADK_MOUNT_SDCARD_FN)(void);
```

Related Data Types and Interfaces

RKADK_RECORD_ATTR_S

2.3.12 RKADK_AOV_ATTR_S

2.3.13 RKADK_RECORD_ATTR_S

Description

Define the recording task attribute structure.

Definition

```
typedef struct {
  RKADK_S32 s32CamID;                                  /* Camera ID */
  RKADK_U32 u32FragKeyFrame;
  RKADK_REC_REQUEST_FILE_NAMES_FN pfnRequestFileNames; /* rec callbak */
  RKADK_REC_EVENT_CALLBACK_FN pfnEventCallback;        /* event callbak */
  RKADK_AOV_ATTR_S stAovAttr;
  RKADK_POST_ISP_ATTR_S *pstPostIspAttr;
  RKADK_MOUMNT_SDCARD_FN pfnMountSdcard;
  RKADK_PIP_ATTR_S stPipAttr[RECORD_FILE_NUM_MAX];
  RKADK_U32 u32GetThumbTime[RECORD_FILE_NUM_MAX];      /* get thumbnail delay
time(ms) when enbale file cache, default: 3000(ms) */
} RKADK_RECORD_ATTR_S;
```

Members

Member Name	Description
s32CamID	Camera ID
u32FragKeyFrame	Whether the video file is sliced into I frames
pfnRequestFileNames	Request file name function pointer
pfnEventCallback	Recording event callback function pointer
stAovAttr	AOV (Always On Video) attributes
pstPostIspAttr	Post AI ISP attributes
pfnMountSdcard	Sdcard mount function pointer
stPipAttr	Picture-in-picture attributes
u32GetThumbTime	Get thumbnail delay time(ms) when enbale file cache

Related data types and interfaces

RKADK_REC_EVENT_CALLBACK_FN

RKADK_REC_REQUEST_FILE_NAMES_FN

RKADK_MOUMNT_SDCARD_FN

RKADK_AOV_ATTR_S

RKADK_POST_ISP_ATTR_S

RKADK_PIP_ATTR_S

RKADK_RECORD_Create

2.3.14 RKADK_MUXER_MANUAL_SPLIT_TYPE_E

Description

Define manual split type.

Definition

```
typedef enum {
  MUXER_PRE_MANUAL_SPLIT,      /* pre manual split type */
  MUXER_NORMAL_MANUAL_SPLIT,   /* normal manual split type */
} RKADK_MUXER_MANUAL_SPLIT_ATTR_S;
```

Members

Member Name	Description
MUXER_PRE_MANUAL_SPLIT	Manually split video files and pre-record
MUXER_NORMAL_MANUAL_SPLIT	Manually split video files

Related data types and interfaces

RKADK_REC_MANUAL_SPLIT_ATTR_S

2.3.15 RKADK_REC_MANUAL_SPLIT_ATTR_S

Description

Define manually split attribute structure.

Definition

```
typedef MUXER_MANUAL_SPLIT_ATTR_S RKADK_REC_MANUAL_SPLIT_ATTR_S;

typedef struct {
  RKADK_MUXER_MANUAL_SPLIT_TYPE_E enManualType;              /* maual split type
*/
  RKADK_U32 u32DurationSec; /* file duration of manual split file */
} MUXER_MANUAL_SPLIT_ATTR_S;
```

Members

Member Name	Description
enManualType	Manually split type
u32DurationSec	Manually split video file duration
Related data types and interfaces	

RKADK_MUXER_MANUAL_SPLIT_TYPE_E

RKADK_RECORD_ManualSplit

2.3.16 FILE_CACHE_ARG

Description

Define the File Cache attribute structure.

Definition

```
typedef struct _FILE_CACHE_ARG {
  const char *sdcard_path;
  int write_cache;  /* write cache size(byte), default 1M */
  int total_cache;  /* total cache size(byte), default 10M */
  FILE_WRITE_THREAD_ARG write_thread_arg;
  FILE_SDCARD_ARG sdcard_arg;
} FILE_CACHE_ARG;
```

Members

Member Name	Description
sdcard_path	SD card mounting path
write_cache	Cache size for each write file
total_cache	Total Cache size
write_thread_arg	Write thread attribute
sdcard_arg	sdcard attribute

Related data types and interfaces

RKADK_RECORD_FileCacheInit

FILE_WRITE_THREAD_ARG

FILE_SDCARD_ARG

2.3.17 FILE_WRITE_THREAD_ARG

Description

Define the write thread attribute structure.

Definition

```
typedef enum _FILE_SCHED_POLICY {
  FILE_SCHED_OTHER = 0,
  FILE_SCHED_BATCH,
  FILE_SCHED_IDLE,
  FILE_SCHED_FIFO,      /* sched_priority[1, 99] */
  FILE_SCHED_RR,        /* sched_priority[1, 99] */
} FILE_SCHED_POLICY;

typedef struct _FILE_WRITE_THREAD_ARG {
  FILE_SCHED_POLICY sched_policy;
  int priority; /* SCHED_OTHER/SCHED_IDLE/SCHED_BATCH inoperative */
} FILE_WRITE_THREAD_ARG;
```

Members

Member Name	Description
sched_policy	Thread schedule policy
priority	Thread priority

Related data types and interfaces

RKADK_RECORD_FileCacheInit

FILE_CACHE_ARG

2.3.18 FILE_SDCARD_ARG

Description

Define the sdcard attribute structure.

Definition

```
typedef struct _FILE_SDCARD_ARG {
    FILE_CACHE_SDCARD_LOCK lock;
    FILE_CACHE_SDCARD_UNLOCK unlock;
    FILE_CACHE_MOUMNT_SDCARD mount_sdcard;
    FILE_CACHE_UMOUMNT_SDCARD umount_sdcard;
} FILE_SDCARD_ARG;
```

Members

Member Name	Description
lock	sdcard operation lock function
unlock	sdcard operation unlock function
mount_sdcard	sdcard mount function
umount_sdcard	sdcard unmount function

Related data types and interfaces

FILE_CACHE_ARG

2.3.19 RKADK_POST_ISP_ATTR_S

Description

Define the post AI ISP attribute structure.

Definition

```
typedef struct {
  AIISP_CALLBACK_FUNC_S stAiIspCallback;      /* post isp callback function */
  const RK_CHAR        *pModelFilePath;       /* post isp model file path   */
  RK_U32                u32FrameBufCnt;       /* RW; frame buffer cnt    */
} RKADK_POST_ISP_ATTR_S;
```

Members

Member Name	Description
stAiIspCallback	AI NR parameter update callback function
pModelFilePath	Post AI ISP model file path
u32FrameBufCnt	frame buffer cnt, default 1

Related data types and interfaces

RKADK_MEDIA_EnablePostIsp

RKADK_MEDIA_SetPostIspAttr

2.3.20 RKADK_AOV_ATTR_S

Description

Define the AOV attribute structure.

Definition

```
typedef struct {
  RKADK_ISP_WAKE_UP_PAUSE_FN pfnSingleFrame;
  RKADK_ISP_WAKE_UP_RESUME_FN pfnMultiFrame;
} RKADK_AOV_ATTR_S;
```

Members

Member Name	Description
pfnSingleFrame	Pointer to the single-frame mode function
pfnMultiFrame	Pointer to the multi-frame mode function

Related Data Types and Interfaces

RKADK_RECORD_Create

2.3.21 RKADK_PIP_ATTR_S

Description

Define the structure for picture-in-picture attributes.

Definition

```
typedef struct {
  RKADK_BOOL bEnablePip;                    /* enable picture-in-picture */
  RKADK_U32 u32AvsGrpId;                    /* AVS group id [0,
AVS_MAX_GRP_NUM)] */
  RKADK_U32 u32AvsBufCnt;                   /* default 2, min value 2 */
  RKADK_U32 u32SubCamId;                    /* subwindow camera id */
  RKADK_STREAM_TYPE_E enSubStreamType;      /* subwindow stream type */
  RKADK_RECT_S stSubRect;                   /* The subwindow is based on the
display area of the main window */
} RKADK_PIP_ATTR_S;
```

Members

Member Name	Description
bEnablePip	Enable picture-in-picture
u32AvsGrpId	AVS Group Id used for picture-in-picture
u32AvsBufCnt	Maximum number of target image buffers for AVS
u32SubCamId	Subwindow Camera Id
enSubStreamType	Subwindow stream type
stSubRect	Display area of the subwindow within the main window

Related Data Types and Interfaces

RKADK_RECORD_Create

RKADK_RECORD_SetPipAttr

3. AOV(Always On Video)

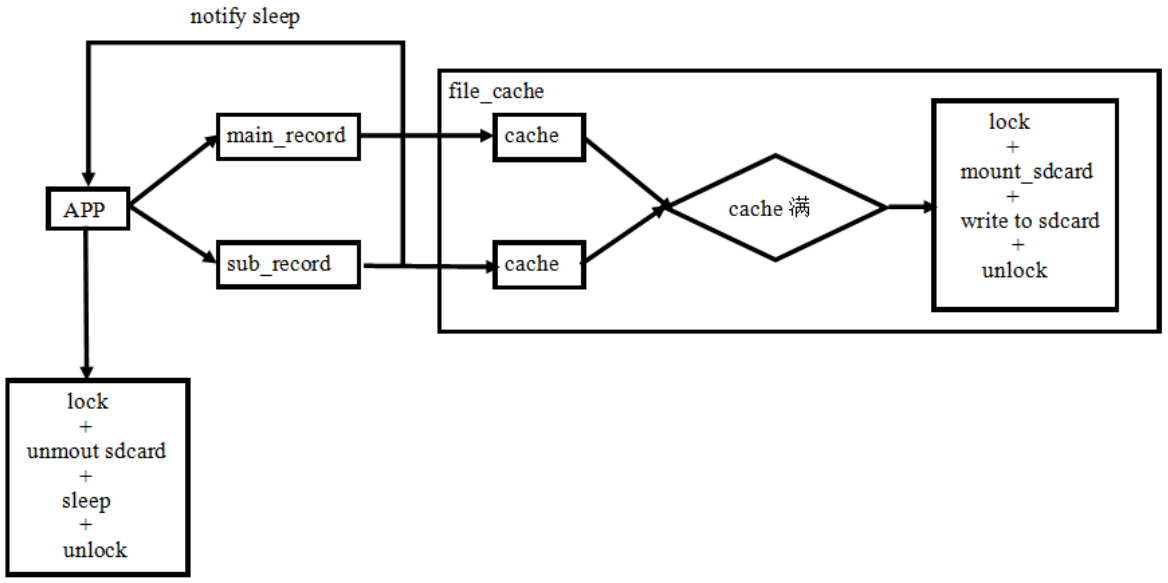
3.1 Overview

AOV (Always On Video) is a low frame rate continuous recording based on sleep&wake-up. It can switch back to the normal frame rate mode when there is an event, such as AI detection, PIR trigger, WIFI wake-up, etc.

It is recommended to enable the file cache function when using AOV. For file cache usage, please refer to the RKADK_RECORD_FileCacheInit and RKADK_RECORD_FileCacheDeInit APIs in the recording chapter of this document.

In order to reduce power consumption in the AOV time-lapse recording mode, the sdcard needs to be powered off after sleeping, and the sdcard must be unmounted before the application goes to sleep. When the file cache is enabled, the file cache will try to mount the sdcard when writing files, and perform file write protection, to ensure that the sdcard will not be unmounted during the file writing process.

The AOV process is shown in the figure below:



3.2 API Reference

3.2.1 RKADK_AOV_Init

Description

AOV initialization.

Grammar

int RKADK_AOV_Init(RKADK_AOV_ARG_S *pstAovAttr);

Parameter

Parameter name	Description	Input/output
pstAovAttr	AOV attribute pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_aov.h

Library file: librkadk.so

Notice

- It Must be used in conjunction with RKADK_AOV_DeInit.
- Repeated initialization is not supported.

Example

rkadk_aov_record_test

Related topic

RKADK_AOV_DeInit

3.2.2 RKADK_AOV_DeInit

Description

AOV deinitialization.

Grammar

int RKADK_AOV_DeInit();

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_aov.h

Library file: librkadk.so

Notice

- It Must be used in conjunction with RKADK_AOV_Init.
- Repeated deinitialization is not supported.

Example

rkadk_aov_record_test

Related topic

RKADK_AOV_Init

3.2.3 RKADK_AOV_SetSuspendTime

Description

Set the AOV scheduled sleep time in ms.

Grammar

int RKADK_AOV_SetSuspendTime(int u32WakeupSuspendTime);

Parameter

Parameter name	Description	Input/output
u32WakeupSuspendTime	Sleep time	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_aov.h

Library file: librkadk.so

Example

rkadk_aov_record_test

3.2.4 RKADK_AOV_EnterSleep

Description

Enter AOV.

Grammar

int RKADK_AOV_EnterSleep();

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_aov.h

Library file: librkadk.so

Example

rkadk_aov_record_test

3.2.5 RKADK_AOV_WakeupLock

Description

Lock AOV related operations.

Grammar

void RKADK_AOV_WakeupLock();

Requirement

Header file: rkadk_aov.h

Library file: librkadk.so

Notice

- Must be used in conjunction with RKADK_AOV_WakeupUnlock.

Example

rkadk_aov_record_test

Related topic

RKADK_AOV_WakeupUnlock

3.2.6 RKADK_AOV_WakeupUnlock

Description

Unlocked AOV related operations.

Grammar

void RKADK_AOV_WakeupUnlock();

Requirement

Header file: rkadk_aov.h

Library file: librkadk.so

Notice

- Must be used in conjunction with RKADK_AOV_WakeupLock.

Example

rkadk_aov_record_test

Related topic

RKADK_AOV_WakeupLock

3.3 Type of data

The AOV module mainly provides the following data types:

RKADK_AOV_ARG_S: AOV attribute structure

RKADK_AOV_NOTIFY_CALLBACK: AOV notification callback function pointer

3.3.1 RKADK_AOV_ARG_S

Definition

Define the AOV attribute structure.

definition

```
typedef struct {
   RKADK_AOV_NOTIFY_CALLBACK pfnNotifyCallback;
} RKADK_AOV_ARG_S;
```

Members

Member Name	Description
pfnNotifyCallback	AOV notification callback function pointer

[Related data types and interfaces]

RKADK_AOV_Init

RKADK_AOV_NOTIFY_CALLBACK

3.3.2 RKADK_AOV_NOTIFY_CALLBACK

Definition

Define the AOV notification callback function pointer.

Definition

```
typedef void (*RKADK_AOV_NOTIFY_CALLBACK)(RKADK_AOV_EVENT_E enEvent, void *msg);
```

Members

Member Name	Description
enEvent	AOV notification event enumeration
msg	Reserve

[Related data types and interfaces]

RKADK_AOV_ARG_S

RKADK_AOV_EVENT_E

3.3.3 RKADK_AOV_EVENT_E

Definition

Defines the AOV notification event enumeration.

definition

```
typedef enum {
   RKADK_AOV_ENTER_SLEEP = 0,
   RKADK_AOV_EVENT_BUTT
} RKADK_AOV_EVENT_E;
```

Members

Member Name	Description
RKADK_AOV_ENTER_SLEEP	Enter AOV sleep

[Related data types and interfaces]

RKADK_AOV_NOTIFY_CALLBACK

4. Taking Photos

4.1 Overview

Provides basic snapshot functionality, offering JPEG encapsulation for capturing photos. The features include:

- Single snapshot
- Multiple snapshots
- JPEG Slice: Encodes VI data into JPEG in segments, which can save memory usage. When enlarging the JPEG, use interpolation algorithms to enlarge VI data in segments, and then encode it into JPEG in segments.

Notice about JPEG Slice:

- Segment height requires alignment to 16 pixels.
- Considering the stitching effect of final images, it is recommended to keep the aspect ratio of the segmented image as consistent as possible with the original images.
- The more segments, the less memory required, but the image generation time increases. Conversely, when the number of segments decreases, the required memory increases, but the image generation speed accelerates; the actual number of segments depends on the actual situation.

4.2 API Reference

4.2.1 RKADK_PHOTO_Init

Description

The photo taking task is initialized.

Syntax

RKADK_S32 RKADK_PHOTO_Init(RKADK_PHOTO_ATTR_S *pstPhotoAttr, RKADK_MW_PTR *ppHandle);

Parameters

Parameter name	Description	Input/output
pstPhotoAttr	Photo task attribute pointer	Input
ppHandle	Created photo task pointer	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_photo.h

Library file: librkadk.so

Notice

- Repeated initialization is not supported.

Example

rkadk_photo_test

See Also

RKADK_PHOTO_DeInit

4.2.2 RKADK_PHOTO_DeInit

Description

Deinitialize the photo taking task.

Syntax

RKADK_S32 RKADK_PHOTO_DeInit(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Pointer to photo taking task	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_photo.h

Library file: librkadk.so

Notice

- Repeated deinitialization is not supported.

Example

rkadk_photo_test

See Also

RKADK_PHOTO_DeInit

4.2.3 RKADK_PHOTO_TakePhoto

Description

Take Photos.

Syntax

RKADK_S32 RKADK_PHOTO_TakePhoto(RKADK_MW_PTR pHandle, RKADK_TAKE_PHOTO_ATTR_S *pstAttr);

Parameters

Parameter name	Description	Input/output
PHandle	Pointer to photo taking task	Input
pstAttr	Photo attributes	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_photo.h

Library file: librkadk.so

Notice

- The RKADK_PHOTO_TakePhoto interface can only be used after the recording task is initialized.

Example

rkadk_photo_test

See Also

No

4.2.4 RKADK_PHOTO_Reset

Description

Reconfigure the photo taking task.

Syntax

RKADK_S32 RKADK_PHOTO_Reset(RKADK_MW_PTR *ppHandle);

Parameters

Parameter name	Description	Input/output
ppHandle	Pointer to photo taking task	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_photo.h

Library file: librkadk.so

Notice

- The RKADK_PHOTO_Reset interface can only be used after initializing the recording task.
- When switching resolution, reset the resolution of the Photo module in the ini file by using the RKADK_PARAM_SetCamParam API.
- RV1126/RV1109 and RK3576 does not support dynamic resolution switching. You need to RKADK_PHOTO_DeInit first, then configure the new resolution, and finally RKADK_PHOTO_Init again.
- If Photo and other modules share the VI channel, both the resolution of Photo and multiplex modules need to be switched simultaneously to avoid VI and VENC resolution mismatch.

Example

rkadk_photo_test

See Also

4.2.5 RKADK_PHOTO_GetThmInJpg

Description

Get thumbnail data from a JPG file.

Syntax

RKADK_S32 RKADK_PHOTO_GetThmInJpg(RKADK_U32 u32CamId, RKADK_CHAR *pszFileName,
RKADK_JPG_THUMB_TYPE_E eThmType,
RKADK_U8 *pu8Buf, RKADK_U32 *pu32Size);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
pszFileName	JPG file path	Input
eThmType	Thumbnail Type	Input
pu8Buf	Input: thumbnail data storage pointer, output: actual thumbnail data	Input/Output
pu32Size	Input: length of pu8Buf , output: length of actual thumbnail data	Input/Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_photo.h

Library file: librkadk.so

Example

rkadk_thumb_test

See Also

No.

4.2.6 RKADK_PHOTO_GetThmInJpgEx

Description

JPG Thumbnail obtaining extended Interface: get thumbnail data from a JPG file, supporting the specification of the output thumbnail type and resolution. It must be used together with RKADK_PHOTO_ThumbBufFree.

Syntax

RKADK_S32 RKADK_PHOTO_GetThmInJpgEx(RKADK_U32 u32CamId, RKADK_CHAR *pszFileName,
RKADK_JPG_THUMB_TYPE_E eThmType,
RKADK_THUMB_ATTR_S *pstThumbAttr);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
pszFileName	JPG file path	Input
eThmType	Thumbnail Type	Input
pstThumbAttr	Thumbnail attribute structure pointer	Input/output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_photo.h

Library file: librkadk.so

Example

rkadk_thumb_test

See Also

RKADK_PHOTO_ThumbBufFree

4.2.7 RKADK_PHOTO_ThumbBufFree

Description

Release the memory requested by RKADK_PHOTO_GetThmInJpgEx. It must be used together with RKADK_PHOTO_GetThmInJpgEx.

Syntax

RKADK_S32 RKADK_PHOTO_ThumbBufFree(RKADK_THUMB_ATTR_S *pstThumbAttr);

Parameters

Parameter name	Description	Input/output
pstThumbAttr	Thumbnail attribute structure pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_photo.h

Library file: librkadk.so

Example

rkadk_thumb_test

See Also

RKADK_PHOTO_GetThmInJpgEx

4.2.8 RKADK_MEDIA_SetVencRotation

Description

Set VENC channel rotation.

Syntax

RKADK_S32 RKADK_MEDIA_SetVencRotation(RKADK_U32 u32CamId, ROTATION_E enRotation, RKADK_STREAM_TYPE_E enStreamType);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
enRotation	Rotation angle	Input
enStreamType	Stream type	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_media_comm.h

Library file: librkadk.so

4.2.9 RKADK_MEDIA_ToggleVencMirror

Description

Set VENC channel Mirror.

Syntax

RKADK_S32 RKADK_MEDIA_ToggleVencMirror(RKADK_U32 u32CamId, RKADK_STREAM_TYPE_E enStrmType, bool mirror);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
enStreamType	Stream type	Input
mirror	Whether to enable mirror	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_media_comm.h

Library file: librkadk.so

4.2.10 RKADK_MEDIA_ToggleVencFlip

Description

Set VENC channel Flip.

Syntax

RKADK_S32 RKADK_MEDIA_ToggleVencFlip(RKADK_U32 u32CamId, RKADK_STREAM_TYPE_E enStrmType, bool flip);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
enStreamType	Stream type	Input
flip	Whether to enable flip	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_media_comm.h

Library file: librkadk.so

4.3 Type of Data

The camera module mainly provides the following data types:

RKADK_PHOTO_TYPE_E: Photo type enumeration

RKADK_PHOTO_SINGLE_ATTR_S: Single snapshot attribute structure

RKADK_PHOTO_MULTIPLE_ATTR_S: Multi snapshots attribute structure

RKADK_PHOTO_RECV_DATA_S: Photo data structure

RKADK_PHOTO_DATA_RECV_FN_PTR: Photo data receiving function pointer

RKADK_TAKE_PHOTO_ATTR_S: Photo attribute structure

RKADK_PHOTO_ATTR_S: Photo task attribute structure

RKADK_JPG_THUMB_TYPE_E: JPG thumbnail type enumeration

RKADK_THUMB_TYPE_E: Output thumbnail type enumeration

RKADK_THUMB_ATTR_S: Thumbnail attribute structure

ROTATION_E: Rotation type enumeration

RKADK_PHOTO_FMT_CHANGE_S: Format conversion parameter structure, used for FBC0/NV16 format conversion

4.3.1 RKADK_PHOTO_TYPE_E

Description

Define the photo type enumeration.

Definition

```
typedef enum {
  RKADK_PHOTO_TYPE_SINGLE = 0,
  RKADK_PHOTO_TYPE_MULTIPLE,
  RKADK_PHOTO_TYPE_LAPSE, // TODO
  RKADK_PHOTO_TYPE_BUTT
} RKADK_PHOTO_TYPE_E;
```

Members

Member Name	Description
RKADK_PHOTO_TYPE_SINGLE	Single snapshot mode
RKADK_PHOTO_TYPE_MULTIPLE	Multi snapshots mode
RKADK_PHOTO_TYPE_LAPSE	Time-lapse photography (Reserved)

Related data types and interfaces

RKADK_TAKE_PHOTO_ATTR_S

4.3.2 RKADK_PHOTO_SINGLE_ATTR_S

Description

Define the single photo attribute structure.

Definition

```
typedef struct {
  // TODO
  RKADK_S32 s32Time_sec;
} RKADK_PHOTO_SINGLE_ATTR_S;
```

Members

Member Name	Description
s32Time_sec	Reserved

Related data types and interfaces

RKADK_TAKE_PHOTO_ATTR_S

4.3.3 RKADK_PHOTO_MULTIPLE_ATTR_S

Description

Define the multi photos attribute structure.

Definition

```
typedef struct {
  /* s32Count is -1 that means continuous photo, larger than 0 that meas photo
   * number */
  RKADK_S32 s32Count;
} RKADK_PHOTO_MULTIPLE_ATTR_S;
```

Members

Member Name	Description
s32Count	Number of continuous photo taking, -1 means continuous photo taking until RKADK_PHOTO_DeInit is called

Related data types and interfaces

RKADK_TAKE_PHOTO_ATTR_S

4.3.4 RKADK_PHOTO_THUMB_ATTR_S

Description

Define the thumbnail parameter attribute structure.

Definition

```
#define RKADK_MPF_LARGE_THUMB_NUM_MAX 2  /* Supports the maximum number of MPF
thumbnails to be generated simultaneously */

typedef struct rkSIZE_S {
  RK_U32 u32Width;              /* Thumbnail width*/
  RK_U32 u32Height;             /* Thumbnail height */
} SIZE_S;

typedef enum {
  RKADK_PHOTO_MPF_SINGLE = 0,    /* Single MPF thumbnail */
  RKADK_PHOTO_MPF_MULTI,         /* Multiple MPF thumbnails */
  RKADK_PHOTO_MPF_BUTT
} RKADK_PHOTO_MPF_MODE_E;

typedef struct {
  RKADK_U8 u8LargeThumbNum;      /* Number of generated MPF thumbnails*/
  SIZE_S astLargeThumbSize[RKADK_MPF_LARGE_THUMB_NUM_MAX];  /* Each thumbnail
resolution */
} RKADK_PHOTO_MPF_CFG_S;

typedef struct {
  RKADK_PHOTO_MPF_MODE_E eMode;  /* MPF thumbnail mode */
  RKADK_PHOTO_MPF_CFG_S sCfg;    /* MPF thumbnail configuration */
} RKADK_PHOTO_MPF_ATTR_S;

typedef struct {
```

```
  RKADK_BOOL bSupportDCF; /* Fixed resolution: 160 * 120 jpg */
  RKADK_PHOTO_MPF_ATTR_S stMPFAttr;
} RKADK_PHOTO_THUMB_ATTR_S;
```

Members

Member Name	Description
bSupportDCF	Whether to generate DCF thumbnails (fixed resolution 160*120)
stMPFAttr	MPF thumbnail attribute parameters

Related data types and interfaces

RKADK_PHOTO_ATTR_S

4.3.5 RKADK_PHOTO_RECV_DATA_S

Description

Define the photo data structure.

Definition

```
typedef struct {
  RKADK_U8 *pu8DataBuf;
  RKADK_U32 u32DataLen;
  RKADK_U32 u32CamId;
  bool bStreamEnd;
  void *userdata;
} RKADK_PHOTO_RECV_DATA_S;
```

Members

Member Name	Description
pu8DataBuf	Data pointer
u32DataLen	Data length
u32CamId	Camera ID
bStreamEnd	Whether the data stream ends, mainly used for JPEG Slice
userdata	User data pointer

Related data types and interfaces

RKADK_PHOTO_DATA_RECV_FN_PTR

4.3.6 RKADK_PHOTO_DATA_RECV_FN_PTR

Description

Define the camera data receiving function pointer.

Definition

```
typedef void (*RKADK_PHOTO_DATA_RECV_FN_PTR)(RKADK_PHOTO_RECV_DATA_S *pstData);
```

Members

Member Name	Description
pstData	Data pointer

Related data types and interfaces

RKADK_PHOTO_RECV_DATA_S

RKADK_PHOTO_ATTR_S

4.3.7 RKADK_TAKE_PHOTO_ATTR_S

Description

Define the photo attribute structure.

Definition

```
typedef struct {
  RKADK_PHOTO_TYPE_E enPhotoType;
  union tagPhotoTypeAttr {
    RKADK_PHOTO_SINGLE_ATTR_S stSingleAttr;
    RKADK_PHOTO_LAPSE_ATTR_S stLapseAttr; // TODO
    RKADK_PHOTO_MULTIPLE_ATTR_S stMultipleAttr;
  } unPhotoTypeAttr;
} RKADK_TAKE_PHOTO_ATTR_S;
```

Members

Member Name	Description
enPhotoType	Photo type
stSingleAttr	Single photo parameter attribute
stMultipleAttr	Multi photos parameter attribute
stLapseAttr	Time-lapse photo parameter attribute (Reserved)

Related data types and interfaces

RKADK_PHOTO_TYPE_E

RKADK_PHOTO_SINGLE_ATTR_S

RKADK_PHOTO_MULTIPLE_ATTR_S

RKADK_PHOTO_TakePhoto

4.3.8 RKADK_PHOTO_FMT_CHANGE_S

Description

Define the format conversion parameter structure, used for FBC0/NV16 format conversion

Definition

```
typedef struct {
  RKADK_U32 u32VencChn;
  RKADK_U32 u32VdecChn;

  //format change venc parameter
  RKADK_U32 u32BitRate;   //default 4M
  RKADK_U32 u32Profile;   //default 77
} RKADK_PHOTO_FMT_CHANGE_S;
```

Members

Member Name	Description
u32VencChn	VENC channel ID used for format conversion
u32VdecChn	VDEC channel ID used for format conversion
u32BitRate	Encoding bitrate, default 4M
u32Profile	Encoding level, default 77

Related Data Types and Interfaces

RKADK_PHOTO_ATTR_S

RKADK_PHOTO_Init

4.3.9 RKADK_PHOTO_ATTR_S

Description

Define the photo task attribute structure.

Definition

```
typedef struct {
  RKADK_U32 u32CamId;
  RKADK_PHOTO_THUMB_ATTR_S stThumbAttr;
  RKADK_PHOTO_DATA_RECV_FN_PTR pfnPhotoDataProc;
  void *userdata;
  RKADK_POST_ISP_ATTR_S *pstPostIspAttr;
  RKADK_PHOTO_FMT_CHANGE_S stFmtChange;
} RKADK_PHOTO_ATTR_S;
```

Members

Member Name	Description
u32CamId	Camera ID
stThumbAttr	Thumbnail parameter attribute
pfnPhotoDataProc	Photo data receiving callback function pointer
userdata	User data pointer
pstPostIspAttr	Post AI ISP attributes
stFmtChange	Format conversion parameters for FBC0/NV16 format conversion

Related data types and interfaces

RKADK_PHOTO_THUMB_ATTR_S

RKADK_PHOTO_DATA_RECV_FN_PTR

RKADK_POST_ISP_ATTR_S

RKADK_PHOTO_FMT_CHANGE_S

RKADK_PHOTO_Init

4.3.10 RKADK_JPG_THUMB_TYPE_E

Description

Defines the JPG thumbnail type enumeration.

Definition

```
typedef enum {
  RKADK_JPG_THUMB_TYPE_DCF,
  RKADK_JPG_THUMB_TYPE_MFP1,
  RKADK_JPG_THUMB_TYPE_MFP2,
  RKADK_JPG_THUMB_TYPE_BUTT
} RKADK_JPG_THUMB_TYPE_E;
```

Members

Member Name	Description
RKADK_JPG_THUMB_TYPE_DCF	DCF thumbnail
RKADK_JPG_THUMB_TYPE_MFP1	MPF1 thumbnail
RKADK_JPG_THUMB_TYPE_MFP2	MPF2 thumbnail

Related data types and interfaces

RKADK_PHOTO_GetThmInJpg

RKADK_PHOTO_GetThmInJpgEx

4.3.11 RKADK_THUMB_TYPE_E

Description

Defines the output thumbnail type enumeration.

Definition

```
typedef enum {
  RKADK_THUMB_TYPE_NV12 = 0,
  RKADK_THUMB_TYPE_JPEG,
  RKADK_THUMB_TYPE_RGB565,
  RKADK_THUMB_TYPE_RGBA8888,
  RKADK_THUMB_TYPE_BGRA8888
} RKADK_THUMB_TYPE_E;
```

Members

Member Name	Description
RKADK_THUMB_TYPE_NV12	Output thumbnails in NV12 format
RKADK_THUMB_TYPE_JPEG	Output thumbnails in JPG format
RKADK_THUMB_TYPE_RGB565	Output thumbnails in RGB565 format
RKADK_THUMB_TYPE_RGBA8888	Output thumbnails in RGBA9888 format
RKADK_THUMB_TYPE_BGRA8888	Output format thumbnails in BGRA8888

Related data types and interfaces

RKADK_THUMB_ATTR_S

RKADK_GetThmInMp4Ex

RKADK_PHOTO_GetThmInJpgEx

4.3.12 RKADK_THUMB_ATTR_S

Description

Define the thumbnail attribute structure.

Definition

```
typedef struct {
  RKADK_THUMB_TYPE_E enType;
  // 4 alignment
  RKADK_U32 u32Width;
  // 2 alignment
  RKADK_U32 u32Height;
  // 4 alignment
  RKADK_U32 u32VirWidth;
  // 2 alignment
  RKADK_U32 u32VirHeight;
  RKADK_U8 *pu8Buf;
  RKADK_U32 u32BufSize;
} RKADK_THUMB_ATTR_S;
```

Members

Member Name	Description
enType	The the desired thumbnail type
u32Width	Enter the desired thumbnail width and output the actual thumbnail width
u32Height	Enter the desired thumbnail height and output the actual thumbnail height
u32VirWidth	Enter the desired virtual width of the thumbnail and output the actual virtual width of the thumbnail
u32VirHeight	Enter the desired virtual height of the thumbnail and output the actual virtual height of the thumbnail
pu8Buf	Thumbnail data pointer
u32BufSize	Thumbnail data length

Related data types and interfaces

RKADK_THUMB_TYPE_E

RKADK_GetThmInMp4Ex

RKADK_PHOTO_GetThmInJpgEx

4.3.13 ROTATION_E

Description

Defines an enumeration of rotation types.

Definition

```
typedef enum rkROTATION_E {
    ROTATION_0        = 0,
    ROTATION_90       = 1,
    ROTATION_180      = 2,
    ROTATION_270      = 3,
    ROTATION_BUTT
} ROTATION_E;
```

Related data types and interfaces

RKADK_MEDIA_SetVencRotation

5. Remote Preview

5.1 Overview

It provides callback interfaces for previewing to obtain Video and Audio information, start and stop VENC, start and stop AENC, and register functions for processing audio and video frame data.

5.2 API Reference

5.2.1 RKADK_STREAM_VideoInit

Description

Initialize Video module: VI, VENC.

Syntax

RKADK_S32 RKADK_STREAM_VideoInit(RKADK_STREAM_VIDEO_ATTR_S *pstVideoAttr, RKADK_MW_PTR *ppHandle);

Parameters

Parameter name	Description	Input/output
pstVideoAttr	Video attribute pointer	Input
ppHandle	Created Video task pointer	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

- Repeated initialization of the Video module is not supported.

Example

rkadk_stream_test

See Also

RKADK_STREAM_VideoDeInit

5.2.2 RKADK_STREAM_VideoDeInit

Description

Deinitialize Video modules: VI, VENC.

Syntax

RKADK_S32 RKADK_STREAM_VideoDeInit(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Video task pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

- Repeated deinitialization is not supported.

Example

rkadk_stream_test

See Also

RKADK_STREAM_VideoInit

5.2.3 RKADK_STREAM_VencStart

Description

Enable VENC.

Syntax

RKADK_S32 RKADK_STREAM_VencStart(RKADK_MW_PTR pHandle, RKADK_S32 s32FrameCnt);

Parameters

Parameter name	Description	Input/output
PHandle	Video task pointer	Input
s32FrameCnt	Specify the number of image frames to be received, -1 means infinite reception until VencStop is called	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

- The RKADK_STREAM_VencStart interface can be called only after the Video module is initialized.
- After calling RKADK_STREAM_VencStart, trigger the VENC data callback function to start receiving data.

Example

rkadk_stream_test

See Also

RKADK_STREAM_VencStop

5.2.4 RKADK_STREAM_VencStop

Description

Disable VENC.

Syntax

RKADK_S32 RKADK_STREAM_VencStop(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Video task pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

None

Example

rkadk_stream_test

See Also

RKADK_STREAM_VencStart

5.2.5 RKADK_STREAM_VideoReset

Description

Reconfigure the stream video..

Syntax

RKADK_S32 RKADK_STREAM_VideoReset(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Video task pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

- The RKADK_STREAM_VideoReset interface can only be used after RKADK_STREAM_VideoInit.

- When switching resolution, frame rate, bit rate, encoding type, or recording type, reset the parameters of the Preview module in the ini file by using the RKADK_PARAM_SetCamParam API.
- RV1126/RV1109 and RK3576 does not support dynamic switching of resolution and encoding type. You need to RKADK_STREAM_VideoDeInit first, then configure the new resolution or encoding type, and finally RKADK_STREAM_VideoInit again.
- If Preview and other modules share the VI channel, both the resolution of Preview and multiplexing modules need to be switched simultaneously to avoid VI and VENC resolution mismatch.

Example

rkadk_stream_test

5.2.6 RKADK_STREAM_GetVideoInfo

Description

Get Video information.

Syntax

RKADK_S32 RKADK_STREAM_GetVideoInfo(RKADK_U32 u32CamId, RKADK_VIDEO_INFO_S *pstVideoInfo);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
pstVideoInfo	Video information structure pointer	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

None

Example

rkadk_stream_test

See Also

None

5.2.7 RKADK_STREAM_AudioInit

Description

Initialize Audio modules: AI, AENC.

Syntax

RKADK_S32 RKADK_STREAM_AudioInit(RKADK_STREAM_AUDIO_ATTR_S *pstAudioAttr, RKADK_MW_PTR *ppHandle);

Parameters

Parameter name	Description	Input/output
pstAudioAttr	Audio attribute pointer	Input
ppHandle	Created Audio task pointer	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

- Repeated initialization of the Audio module is not supported.

Example

rkadk_stream_test

See Also

RKADK_STREAM_AudioDeInit

5.2.8 RKADK_STREAM_AudioDeInit

Description

Deinitialize Audio modules: AI, AENC.

Syntax

RKADK_S32 RKADK_STREAM_AudioDeInit(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Audio task pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

None

Example

rkadk_stream_test

See Also

RKADK_STREAM_AudioInit

5.2.9 RKADK_STREAM_AencStart

Description

Enable AENC.

Syntax

RKADK_S32 RKADK_STREAM_AencStart(RKADK_MW_PTR *pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Audio task pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

- The RKADK_STREAM_AencStart interface can be called only after the Audio module is initialized.
- After calling RKADK_STREAM_AencStart, trigger the AENC data callback function to start receiving data.

Example

rkadk_stream_test

See Also

RKADK_STREAM_AencStop

5.2.10 RKADK_STREAM_AencStop

Description

Disable AENC.

Syntax

RKADK_S32 RKADK_STREAM_AencStop(RKADK_MW_PTR *pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Audio task pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

None

Example

rkadk_stream_test

See Also

RKADK_STREAM_AencStart

5.2.11 RKADK_STREAM_GetAudioInfo

Description

Get Audio information.

Syntax

RKADK_S32 RKADK_STREAM_GetAudioInfo(RKADK_MW_PTR *pHandle, RKADK_AUDIO_INFO_S *pstAudioInfo);

Parameters

Parameter name	Description	Input/output
PHandle	Audio task pointer	Input
pstAudioInfo	Audio information structure pointer	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_stream.h

Library file: librkadk.so

Notice

None

Example

rkadk_stream_test

See Also

None

5.3 Type of Data

The playback module mainly provides the following data types:

RKADK_CODEC_TYPE_E: Encoding format enumeration

RKADK_VENC_DATA_PROC_FUNC: VENC data callback function pointer

RKADK_VIDEO_STREAM_S: Video data stream structure

RKADK_VENC_DATA_PACK_S: VENC data packet structure

RKADK_VENC_DATA_TYPE_S: VENC packet type

RKADK_VIDEO_INFO_S: Video information structure

RKADK_STREAM_VIDEO_ATTR_S: Video task attribute structure

RKADK_AUDIO_DATA_PROC_FUNC: Audio data callback function pointer

RKADK_AUDIO_STREAM_S: Audio data structure

RKADK_AUDIO_INFO_S: Audio information structure

RKADK_STREAM_AUDIO_ATTR_S: Audio task attribute structure

5.3.1 RKADK_CODEC_TYPE_E

Description

Define the encoding format enumeration type. Audio encoding does not support AAC by default.

Definition

```
typedef enum {
  //Video
  RKADK_CODEC_TYPE_H264 = 0,
  RKADK_CODEC_TYPE_H265,
  RKADK_CODEC_TYPE_MJPEG,
  RKADK_CODEC_TYPE_JPEG,

  //Audio
  RKADK_CODEC_TYPE_G711A,
  RKADK_CODEC_TYPE_G711U,
  RKADK_CODEC_TYPE_G726,
  RKADK_CODEC_TYPE_MP2,
  RKADK_CODEC_TYPE_MP3,
  RKADK_CODEC_TYPE_ACC,
  RKADK_CODEC_TYPE_PCM,
  RKADK_CODEC_TYPE_BUTT
} RKADK_CODEC_TYPE_E;
```

5.3.2 RKADK_VENC_DATA_PROC_FUNC

Description

Define the VENC data callback function pointer.

Definition

```
typedef RKADK_S32 (*RKADK_VENC_DATA_PROC_FUNC)(RKADK_VIDEO_STREAM_S
*pVStreamData);
```

Related data types and interfaces

RKADK_VIDEO_STREAM_S

5.3.3 RKADK_VIDEO_STREAM_S

Description

Define the Video stream structure.

Definition

```
typedef struct {
  RKADK_VENC_DATA_PACK_S astPack; /* stream pack attribute */
  RKADK_U32 u32Seq;               /* the list number of stream */
  RKADK_BOOL bEndOfStream;        /* frame end flag */
  RKADK_U32 u32CamId;
} RKADK_VIDEO_STREAM_S;
```

Members

Member Name	Description
astPack	Packet structure
u32Seq	Packet sequence ID
bEndOfStream	Reserved
u32CamId	Camera ID

Related data types and interfaces

RKADK_VENC_DATA_PACK_S

RKADK_VENC_DATA_PROC_FUNC

5.3.4 RKADK_VENC_DATA_PACK_S

Description

Define the VENC packet structure.

Definition

```
typedef struct {
  RKADK_U8 *apu8Addr;                /* the virtual address of stream */
  RKADK_U32 au32Len;                 /* the length of stream */
  RKADK_U64 u64PTS;                  /* time stamp */
  RKADK_VENC_DATA_TYPE_S stDataType; /* the type of stream */
} RKADK_VENC_DATA_PACK_S;
```

Members

Member Name	Description
apu8Addr	Data pointer
au32Len	Data length
u64PTS	Timestamp
stDataType	Data type

Related data types and interfaces

RKADK_VENC_DATA_TYPE_S

RKADK_VIDEO_STREAM_S

5.3.5 RKADK_VENC_DATA_TYPE_S

Description

Define VENC packet type.

Definition

```
/* the nalu type of H264 */
typedef enum {
  RKADK_H264E_NALU_BSLICE = 0,    /* B SLICE types */
  RKADK_H264E_NALU_PSLICE = 1,    /* P SLICE types */
  RKADK_H264E_NALU_ISLICE = 2,    /* I SLICE types */
  RKADK_H264E_NALU_IDRSLICE = 5, /* IDR SLICE types */
  RKADK_H264E_NALU_SEI = 6,       /* SEI types */
  RKADK_H264E_NALU_SPS = 7,       /* SPS types */
  RKADK_H264E_NALU_PPS = 8,       /* PPS types */
  RKADK_H264E_NALU_BUTT
} RKADK_H264E_NALU_TYPE_E;

/* the nalu type of H265 */
typedef enum {
  RKADK_H265E_NALU_BSLICE = 0,     /* B SLICE types */
  RKADK_H265E_NALU_PSLICE = 1,     /* P SLICE types */
  RKADK_H265E_NALU_ISLICE = 2,     /* I SLICE types */
  RKADK_H265E_NALU_IDRSLICE = 19, /* IDR SLICE types */
  RKADK_H265E_NALU_VPS = 32,       /* VPS types */
  RKADK_H265E_NALU_SPS = 33,       /* SPS types */
  RKADK_H265E_NALU_PPS = 34,       /* PPS types */
  RKADK_H265E_NALU_SEI = 39,       /* SEI types */
  RKADK_H265E_NALU_BUTT
} RKADK_H265E_NALU_TYPE_E;

typedef struct {
  RKADK_CODEC_TYPE_E enPayloadType;      /* H.264/H.265/JPEG/MJPEG */
  union {
    RKADK_H264E_NALU_TYPE_E enH264EType; /* H264E NALU types */
    RKADK_H265E_NALU_TYPE_E enH265EType; /* H265E NALU types */
    RKADK_JPEGE_PACK_TYPE_E enJPEGEType; /* TODO: JPEGE PACK types*/
```

```
  };
} RKADK_VENC_DATA_TYPE_S;
```

Members

Member Name	Description
enPayloadType	Encoding type
enH264EType	H264 encoded packet type
enH265EType	H265 encoded packet type
enJPEGEType	Reserved

Related data types and interfaces

RKADK_CODEC_TYPE_E

RKADK_VENC_DATA_PACK_S

5.3.6 RKADK_VIDEO_INFO_S

Description

Define Video information structure.

Definition

```
typedef struct {
  RKADK_CODEC_TYPE_E enCodecType;
  RKADK_U32 u32Width;
  RKADK_U32 u32Height;
  RKADK_U32 u32BitRate;
  RKADK_U32 u32FrameRate;
  RKADK_U32 u32Gop;
} RKADK_VIDEO_INFO_S;
```

Members

Member Name	Description
enPayloadType	Encoding type
u32Width	Resolution width
u32Height	Resolution height
u32BitRate	Bit rate
u32FrameRate	Frame rate
u32Gop	I frame interval

Related data types and interfaces

RKADK_CODEC_TYPE_E

RKADK_STREAM_GetVideoInfo

5.3.7 RKADK_STREAM_VIDEO_ATTR_S

Description

Define the Video task attribute structure.

Definition

```
typedef struct {
  RKADK_U32 u32CamId;
  RKADK_VENC_DATA_PROC_FUNC pfnDataCB;
} RKADK_STREAM_VIDEO_ATTR_S;
```

Members

Member Name	Description
u32CamId	Camera ID
pfnDataCB	Video data output callback function

Related data types and interfaces

RKADK_VENC_DATA_PROC_FUNC

RKADK_STREAM_VideoInit

5.3.8 RKADK_AUDIO_DATA_PROC_FUNC

Description

Define the AENC data callback function pointer.

Definition

```
typedef RKADK_S32 (*RKADK_AUDIO_DATA_PROC_FUNC)(RKADK_AUDIO_STREAM_S
*pAStreamData);
```

Related data types and interfaces

RKADK_AUDIO_STREAM_S

5.3.9 RKADK_AUDIO_STREAM_S

Description

Define the Audio data flow structure.

Definition

```
typedef struct {
  RKADK_U8 *pStream;      /* the virtual address of stream */
  RKADK_U32 u32Len;       /* stream lenth, by bytes */
  RKADK_U64 u64TimeStamp; /* frame time stamp */
  RKADK_U32 u32Seq;       /* frame seq, if stream is not a valid frame,u32Seq is
0 */
  RKADK_CODEC_TYPE_E enType;
} RKADK_AUDIO_STREAM_S;
```

Members

Member Name	Description
pStream	Data pointer
u32Len	Data length
u64TimeStamp	Timestamp
u32Seq	Serial number
enType	Audio data encoding type

Related data types and interfaces

RKADK_AUDIO_DATA_PROC_FUNC

5.3.10 RKADK_AUDIO_INFO_S

Description

Define the Audio information structure.

Definition

```
typedef struct {
  RKADK_CODEC_TYPE_E enCodecType;
  RKADK_U32 u32ChnCnt;
  RKADK_U32 u32SampleRate;
  RKADK_U32 u32AvgBytesPerSec;
  RKADK_U32 u32SamplesPerFrame;
  RKADK_U16 u16SampleBitWidth;
} RKADK_AUDIO_INFO_S;
```

Members

Member Name	Description
enPayloadType	Encoding type
u32ChnCntt	Number of channels
u32SampleRate	Sampling rate
u32AvgBytesPerSec	Byte rate
u32SamplesPerFrame	Number of samples per frame
u16SampleBitWidth	Number of bits per sample

Related data types and interfaces

RKADK_CODEC_TYPE_E

RKADK_STREAM_GetAudioInfo

5.3.11 RKADK_STREAM_AUDIO_ATTR_S

Description

Define the Audio task attribute structure.

Definition

```
typedef struct {
  RKADK_U32 u32CamId;
  RKADK_CODEC_TYPE_E enCodecType;
  RKADK_AUDIO_DATA_PROC_FUNC pfnPcmDataCB;
  RKADK_AUDIO_DATA_PROC_FUNC pfnAencDataCB;
} RKADK_STREAM_AUDIO_ATTR_S;
```

Members

Member Name	Description
u32CamId	Camera ID
enCodecType	Encoding type
pfnPcmDataCB	PCM data output callback function
pfnAencDataCB	Audio encoded data output callback function

Related data types and interfaces

RKADK_AUDIO_DATA_PROC_FUNC

RKADK_STREAM_AudioInit

6. Player

6.1 Overview

It provides local audio and video files, RTSP network stream playback functions, and supports basic playback control operations: play, pause, seek, and screen snapshot.

The Player function is compatible with RV1109/RV1126, RV1103/RV1103, RK3308, and RK3506 platforms. RV1109/RV1126 uses hardware decoding, and other platforms use software decoding.

6.2 API Reference

6.2.1 RKADK_PLAYER_Create

Description

Create a player.

Syntax

RKADK_S32 RKADK_PLAYER_Create(RKADK_MW_PTR *ppPlayer, RKADK_PLAYER_CFG_S *pstPlayCfg);

Parameters

Parameter name	Description	Input/output
ppPlayer	Created player pointer	Output
pstPlayCfg	Player Properties	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

- Repeated creation of the same player is not supported.

Example

rkadk_player_test

See Also

RKADK_PLAYER_Destroy

6.2.2 RKADK_PLAYER_Destroy

Description

Destroy the player.

Syntax

RKADK_S32 RKADK_PLAYER_Destroy(RKADK_MW_PTR pPlayer);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

- Repeated destruction of the same player is not supported.

Example

rkadk_player_test

See Also

RKADK_PLAYER_Create

6.2.3 RKADK_PLAYER_SetDataSource

Description

Set the path of the file to be played. When Player enables the third-party demuxer library, RKADK_PLAYER_SetDataParam should be used.

Syntax

RKADK_S32 RKADK_PLAYER_SetDataSource(RKADK_MW_PTR pPlayer, const RKADK_CHAR *pszfilePath);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input
pszfilePath	The path of the file to be played	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

- The RKADK_PLAYER_SetDataSource interface can be called only after the player is created.

Example

rkadk_player_test

See Also

None

6.2.4 RKADK_PLAYER_SetDataParam

Description

Set the audio and video parameters of the file to be played. Use this interface when Player enables the third-party demuxer library, otherwise use RKADK_PLAYER_SetDataSource.

Syntax

RKADK_S32 RKADK_PLAYER_SetDataParam(RKADK_MW_PTR pPlayer, RKADK_PLAYER_DATA_PARAM_S *pstDataParam);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input
pstDataParam	Audio and video parameters of the file to be played	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

- The RKADK_PLAYER_SetDataParam interface can be called only after the player is created.

Example

rkadk_player_test

See Also

RKADK_PLAYER_SendAudioPacket

RKADK_PLAYER_SendVideoPacket

6.2.5 RKADK_PLAYER_Prepare

Description

Ready to play.

Syntax

RKADK_S32 RKADK_PLAYER_Prepare(RKADK_MW_PTR pPlayer);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

- The RKADK_PLAYER_Prepare interface can only be called after creating the player and setting the playback path.

Example

rkadk_player_test

See Also

None

6.2.6 RKADK_PLAYER_GetCurrentPosition

Description

Get the current playback progress.

Syntax

RKADK_S64 RKADK_PLAYER_GetCurrentPosition(RKADK_MW_PTR pPlayer);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input

Return value

Return value	Description
Playback progress, unit ms	Success
-1	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

None

Example

rkadk_player_test

See Also

None

6.2.7 RKADK_PLAYER_Play

Description

Start playing.

Syntax

RKADK_S32 RKADK_PLAYER_Play(RKADK_MW_PTR pPlayer);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

- The RKADK_PLAYER_Play interface can be called only after calling RKADK_PLAYER_Prepare.

Example

rkadk_player_test

See Also

RKADK_PLAYER_Stop

6.2.8 RKADK_PLAYER_Stop

Description

Stop playback and release resources.

Syntax

RKADK_S32 RKADK_PLAYER_Stop(RKADK_MW_PTR pPlayer);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

None

Example

rkadk_player_test

See Also

RKADK_PLAYER_Play

6.2.9 RKADK_PLAYER_Pause

Description

Pause playback.

Syntax

RKADK_S32 RKADK_PLAYER_Pause(RKADK_MW_PTR pPlayer);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

None

Example

rkadk_player_test

See Also

RKADK_PLAYER_Play

6.2.10 RKADK_PLAYER_Seek

Description

Seek.

Syntax

RKADK_S32 RKADK_PLAYER_Seek(RKADK_MW_PTR pPlayer, RKADK_S64 s64TimeInMs);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input
s64TimeInMs	Seek duration	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

- The RKADK_PLAYER_Seek interface can be called only after calling RKADK_PLAYER_Play.

Example

rkadk_player_test

See Also

RKADK_PLAYER_Play

6.2.11 RKADK_PLAYER_GetPlayStatus

Description

Get the current playback status.

Syntax

RKADK_S32 RKADK_PLAYER_GetPlayStatus(RKADK_MW_PTR pPlayer, RKADK_PLAYER_STATE_E *penState);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input
penState	Current playback state	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

None

Example

rkadk_player_test

See Also

None

6.2.12 RKADK_PLAYER_GetDuration

Description

Get the duration of current playing file.

Syntax

RKADK_S32 RKADK_PLAYER_GetDuration(RKADK_MW_PTR pPlayer, RKADK_U32 *pDuration);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input
pDuration	The duration of current playing file, unit is ms	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

None

Example

rkadk_player_test

See Also

None

6.2.13 RKADK_PLAYER_Snapshot

Description

Player screenshot, calling this interface will encode the currently displayed picture into JPEG data. The application can obtain the generated JPEG data by registering the RKADK_PPLAYER_SNAPSHOT_RECV_FN callback.

Syntax

RKADK_S32 RKADK_PLAYER_Snapshot(RKADK_MW_PTR pPlayer);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Example

rkadk_player_test

See Also

None

6.2.14 RKADK_PLAYER_SendAudioPacket

Description

When Player enables the third-party demuxer library, this interface is used to send de-capsulated audio data to Player for decoding and playback.

Syntax

RKADK_S32 RKADK_PLAYER_SendAudioPacket(RKADK_MW_PTR pPlayer, RKADK_PLAYER_PACKET *pstPacket);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input
pstPacket	sent packet	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

- Before using this interface, you need to call RKADK_PLAYER_SetDataParam to set audio and video related parameters.

Example

rkadk_player_test

See Also

RKADK_PLAYER_SendVideoPacket

RKADK_PLAYER_SetDataParam

6.2.15 RKADK_PLAYER_SendVideoPacket

Description

When Player enables the third-party demuxer library, this interface is used to send de-capsulated video data to Player for decoding and playback.

Syntax

RKADK_S32 RKADK_PLAYER_SendVideoPacket(RKADK_MW_PTR pPlayer, RKADK_PLAYER_PACKET *pstPacket);

Parameters

Parameter name	Description	Input/output
pPlayer	Player pointer	Input
pstPacket	sent packet	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_player.h

Library file: librkadk.so

Notice

- Before using this interface, you need to call RKADK_PLAYER_SetDataParam to set audio and video related parameters.

Example

rkadk_player_test

See Also

RKADK_PLAYER_SendAudioPacket

RKADK_PLAYER_SetDataParam

6.2.16 RKADK_PLAYER_GetSendFrameNum

Description

Retrieve the number of frames that have been successfully decoded.

Syntax

RKADK_S32 RKADK_PLAYER_GetSendFrameNum(RKADK_MW_PTR pPlayer);

Parameters

Parameter Name	Description	Input/Output
pPlayer	Player pointer	Input

Return Value

Return Value	Description
Frame count	Success
-1	Failure

Requirements

Header file: rkadk_player.h

Library file: librkadk.so

Note

- The count will be reset after Pause/Stop.

Example

rkadk_player_test

Related Topics

None

6.2.17 RKADK_PLAYER_SetVdecWaterline

Description

Set the VDEC waterline.

Syntax

RKADK_S32 RKADK_PLAYER_SetVdecWaterline(RKADK_MW_PTR pPlayer, RKADK_U32 u32VdecWaterline);

Parameters

Parameter Name	Description	Input/Output
pPlayer	Player pointer	Input
u32VdecWaterline	Waterline value	Input

Return Value

Return Value	Description
0	Success
Other value	Failure

Requirements

Header file: rkadk_player.h

Library file: librkadk.so

Note

- After setting the VDEC waterline, playback will only start when the VDEC buffer reaches the waterline value.

Example

rkadk_player_test

Related Topics

None

6.2.18 RKADK_PLAYER_SetAoVolume

Description

Set playback volume.

Syntax

RKADK_S32 RKADK_PLAYER_SetAoVolume(RKADK_MW_PTR pPlayer, RKADK_S32 s32Volume);

Parameters

Parameter Name	Description	Input/Output
pPlayer	Player pointer	Output
s32Volume	Volume value	Input

Return Values

Return Value	Description
0	Success
Other value	Failure

Requirements

Header file: rkadk_player.h

Library file: librkadk.so

Notes

None

Example

rkadk_player_test

Related Topics

None

6.3 Type of Data

The playback module mainly provides the following data types:

RKADK_PLAYER_EVENT_E: Playback event enumeration type

RKADK_PLAYER_EVENT_FN: Playback event callback function pointer

RKADK_PLAYER_CFG_S: Player attribute structure

RKADK_VO_FORMAT_E: Image pixel format enumeration type

RKADK_VO_INTF_TYPE_E: Display interface enumeration type

RKADK_PLAYER_FRAME_INFO_S: Image information structure

RKADK_PLAYER_STATE_E: Playback status enumeration type

RKADK_PLAYER_DATA_PARAM_S: Audio and video parameter structure

RKADK_PLAYER_PACKET: Player data packet structure

RKADK_PLAYER_RTSP_CFG_S: RTSP attribute structure

RKADK_PLAYER_VDEC_CFG_S: VDEC attribute structure

RKADK_PLAYER_SNAPSHOT_CFG_S: Screenshot attribute structure

RKADK_PLAYER_SNAPSHOT_S: Screenshot data structure

RKADK_PPLAYER_SNAPSHOT_RECV_FN: Screenshot data callback function pointer

6.3.1 RKADK_PLAYER_EVENT_E

Description

Defines the playback event enumeration type.

Definition

```
typedef enum {
  RKADK_PLAYER_EVENT_STATE_CHANGED = 0x0,
  RKADK_PLAYER_EVENT_PREPARED,
  RKADK_PLAYER_EVENT_PLAY,
  RKADK_PLAYER_EVENT_PAUSED,
  RKADK_PLAYER_EVENT_STOPPED,
  RKADK_PLAYER_EVENT_EOF,
  RKADK_PLAYER_EVENT_SOF,
  RKADK_PLAYER_EVENT_SEEK_END,
  RKADK_PLAYER_EVENT_ERROR,
  RKADK_PLAYER_EVENT_BUTT
} RKADK_PLAYER_EVENT_E;
```

Members

Member Name	Description
RKADK_PLAYER_EVENT_STATE_CHANGED	Status change (Reserved)
RKADK_PLAYER_EVENT_PREPARED	The `Prepared` is completed
RKADK_PLAYER_EVENT_PLAY	Start playing
RKADK_PLAYER_EVENT_PAUSED	Pause playback (Reserved)
RKADK_PLAYER_EVENT_STOPPED	Stop playing
RKADK_PLAYER_EVENT_EOF	End of playback
RKADK_PLAYER_EVENT_SOF	Reserved
RKADK_PLAYER_EVENT_SEEK_END	The `Seek` is completed (Reserved)

Related data types and interfaces

RKADK_PLAYER_EVENT_FN

6.3.2 RKADK_PLAYER_EVENT_FN

Description

Define the playback event callback function pointer.

Definition

```
typedef RKADK_VOID (*RKADK_PLAYER_EVENT_FN)(RKADK_MW_PTR pPlayer,
RKADK_PLAYER_EVENT_E enEvent, RKADK_VOID *pData);
```

Members

Member Name	Description
pPlayer	Player pointer
enEvent	event type
pData	Event related parameters

Related data types and interfaces

RKADK_PLAYER_EVENT_E

RKADK_PLAYER_CFG_S

6.3.3 RKADK_PLAYER_CFG_S

Description

Define the player attribute structure.

Definition

```
typedef struct {
  RKADK_BOOL bEnableVideo;
  RKADK_BOOL bEnableAudio;
  RKADK_BOOL bEnableThirdDemuxer;
  RKADK_PLAYER_FRAME_INFO_S stFrmInfo;
  RKADK_PLAYER_RTSP_CFG_S stRtspCfg;
  RKADK_PLAYER_VDEC_CFG_S stVdecCfg;
  RKADK_PLAYER_SNAPSHOT_CFG_S stSnapshotCfg;
  RKADK_BOOL bEnableBlackBackground;
  RKADK_PLAYER_EVENT_FN pfnPlayerCallback;
} RKADK_PLAYER_CFG_S;
```

Members

Member Name	Description
bEnableVideo	Enable video playback
bEnableAudio	Enable audio playback
bEnableThirdDemuxer	Enable third-party demuxer library
pfnPlayerCallback	Player event callback function pointer
stFrmInfo	Define image information
stRtspCfg	RTSP attribute
stVdecCfg	VDEC attribute
stSnapshotCfg	Screenshot attribute
bEnableBlackBackground	Whether the screen goes black after playing

Related data types and interfaces

RKADK_PLAYER_EVENT_FN

RKADK_PLAYER_FRAME_INFO_S

RKADK_PLAYER_RTSP_CFG_S

RKADK_PLAYER_VDEC_CFG_S

RKADK_PLAYER_SNAPSHOT_CFG_S

RKADK_PLAYER_Create

6.3.4 RKADK_VO_FORMAT_E

Description

Defines the image pixel format enumeration type.

Definition

```
typedef enum {
  VO_FORMAT_ARGB8888 = 0,
  VO_FORMAT_ABGR8888,
  VO_FORMAT_RGB888,
  VO_FORMAT_BGR888,
  VO_FORMAT_ARGB1555,
  VO_FORMAT_ABGR1555,
  VO_FORMAT_RGB565,
  VO_FORMAT_RGB444,
  VO_FORMAT_NV12,
  VO_FORMAT_NV21
} RKADK_VO_FORMAT_E;
```

Related data types and interfaces

RKADK_PLAYER_FRAME_INFO_S

6.3.5 RKADK_VO_INTF_TYPE_E

Description

Defines the display interface enumeration type.

Definition

```
typedef enum {
  DISPLAY_TYPE_HDMI = 0,
  DISPLAY_TYPE_EDP,
  DISPLAY_TYPE_VGA,
  DISPLAY_TYPE_DP,
  DISPLAY_TYPE_HDMI_EDP,
  DISPLAY_TYPE_MIPI,
  DISPLAY_TYPE_DEFAULT,
} RKADK_VO_INTF_TYPE_E;
```

Members

Member Name	Description
DISPLAY_TYPE_HDMI	The display interface is HDMI
DISPLAY_TYPE_EDP	The display interface is EDP
DISPLAY_TYPE_VGA	The display interface is VGA
DISPLAY_TYPE_MIPI	The display interface is MIPI
DISPLAY_TYPE_DP	The display interface is DP
DISPLAY_TYPE_HDMI_EDP	The display interface is HDMI | EDP
DISPLAY_TYPE_DEFAULT	Internal detection, display interface is the actual connected hardware

Related data types and interfaces

RKADK_PLAYER_FRAME_INFO_S

6.3.6 RKADK_VO_SPLICE_MODE_E

Description

Defines the layer splicing method enumeration type.

Definition

```
typedef enum {
  SPLICE_MODE_RGA = 0,
  SPLICE_MODE_GPU,
  SPLICE_MODE_BYPASS
} RKADK_VO_INTF_TYPE_E;
```

Members

Member Name	Description
SPLICE_MODE_RGA	RGA splicing
SPLICE_MODE_GPU	GPU splicing
SPLICE_MODE_BYPASS	BYPASS, no splicing

Related data types and interfaces

RKADK_PLAYER_FRAME_INFO_S

6.3.7 RKADK_PLAYER_FRAME_INFO_S

Description

Define the image information structure.

Definition

```
typedef struct {
  RKADK_U32 u32FrmInfoX;
  RKADK_U32 u32FrmInfoY;
  RKADK_U32 u32DispWidth;
  RKADK_U32 u32DispHeight;
  RKADK_U32 u32ImgWidth;
  RKADK_U32 u32ImgHeight;
  RKADK_U32 u32VoLay;
  RKADK_U32 u32VoDev;
  RKADK_U32 u32VoChn;
  RKADK_U32 u32BorderColor;
  RKADK_U32 u32BorderTopWidth;
  RKADK_U32 u32BorderBottomWidth;
  RKADK_U32 u32BorderLeftWidth;
  RKADK_U32 u32BorderRightWidth;
  RKADK_BOOL bMirror;
  RKADK_BOOL bFlip;
  RKADK_U32 u32Rotation; //0: 0, 1: 90, 2: 180, 3: 270
  RKADK_VO_FORMAT_E u32VoFormat;
  RKADK_VO_INTF_TYPE_E u32EnIntfType;
  RKADK_VO_INTF_SYNC_E enIntfSync;
  RKADK_VO_SYNC_INFO_S stSyncInfo;
  RKADK_VO_SPLICE_MODE_E enVoSpliceMode;
} RKADK_PLAYER_FRAMEINFO_S;
```

Members

Member Name	Description
u32FrmInfoX	The x coordinate of the layer display area
u32FrmInfoY	The y coordinate of the layer display area
u32DispWidth	The width of the layer display area
u32DispHeight	The height of the layer display area
u32ImgWidth	Layer image width
u32ImgHeight	Layer image height
u32VoLay	Video output video layer number
u32VoDev	Display output device number
u32VoChn	Video output channel number, value range: [0, VO_MAX_CHN_NUM(128)]
u32BorderColor	Video output channel border attribute: color (Reserved)
u32BorderTopWidth	Video output channel border attribute: top border width (Reserved)
u32BorderBottomWidth	Video output channel border attribute: bottom border width (Reserved)
u32BorderLeftWidth	Video output channel border attribute: left border width (Reserved)
u32BorderRightWidth	Video output channel border attribute: right border width (Reserved)
bMirror	Enable mirror
bFlip	Enable flip
u32Rotation	Rotation, values: [0: 0, 1: 90, 2: 180, 3: 270]
u32VoFormat	Define image pixel format
u32EnIntfType	Display interface type
enIntfSync	Screen interface synchronization mode
stSyncInfo	Screen attribute structure
enVoSpliceMode	Layer splice mode

Notice

- For detailed video output related attributes, please refer to the VO chapter of the Rockit document `Rockchip_Developer_Guide_MPI.pdf`.

Related data types and interfaces

RKADK_VO_FORMAT_E

RKADK_VO_INTF_TYPE_E

RKADK_VO_SPLICE_MODE_E

RKADK_PLAYER_Create

6.3.8 RKADK_PLAYER_STATE_E

Description

Defines the playback status enumeration type.

Definition

```
typedef enum {
  RKADK_PLAYER_STATE_IDLE = 0, /* The player state before init */
  RKADK_PLAYER_STATE_INIT,     /* The player is in the initial state. It changes
                                  to the initial state after being SetDataSource
*/
  RKADK_PLAYER_STATE_PREPARED, /* The player is in the prepared state */
  RKADK_PLAYER_STATE_PLAY,     /* The player is in the playing state */
  RKADK_PLAYER_STATE_TPLAY,    /* The player is in the trick playing state,
Reserved */
  RKADK_PLAYER_STATE_PAUSE,    /* The player is in the pause state */
  RKADK_PLAYER_STATE_ERR,      /* The player is in the err state */
  RKADK_PLAYER_STATE_BUTT
} RKADK_PLAYER_STATE_E;
```

Related data types and interfaces

RKADK_PLAYER_GetPlayStatus

6.3.9 RKADK_PLAYER_SNAPSHOT_S

Description

Define the screenshot data structure.

Definition

```
typedef struct {
  RKADK_U32 u32Width;
  RKADK_U32 u32Height;
  RKADK_U32 u32DataLen;
  RKADK_U8 *pu8DataBuf;
} RKADK_PLAYER_SNAPSHOT_S;
```

Members

Member Name	Description
u32Width	The width of the screenshot
u32Height	The height of the screenshot
u32DataLen	The data length of the screenshot
pu8DataBuf	The data pointer of the screenshot

Related data types and interfaces

RKADK_PPLAYER_SNAPSHOT_RECV_FN

6.3.10 RKADK_PPLAYER_SNAPSHOT_RECV_FN

Description

Defines the screenshot data callback function pointer.

Definition

```
typedef void (*RKADK_PPLAYER_SNAPSHOT_RECV_FN)(RKADK_PLAYER_SNAPSHOT_S *pstData);
```

Members

Member Name	Description
pstData	The data pointer of the screenshot

Related data types and interfaces

RKADK_PLAYER_SNAPSHOT_S

RKADK_PLAYER_SNAPSHOT_CFG_S

6.3.11 RKADK_PLAYER_SNAPSHOT_CFG_S

Description

Define the screenshot attribute structure.

Definition

```
typedef struct {
  RKADK_U32 u32VencChn;
  RKADK_U32 u32MaxWidth;    //Support snapshot max width, default 4096
  RKADK_U32 u32MaxHeight;   //Support snapshot max height, default 4096
  RKADK_PPLAYER_SNAPSHOT_RECV_FN pfnDataCallback;
} RKADK_PLAYER_SNAPSHOT_CFG_S;
```

Members

Member Name	Description
u32VencChn	JPEG encoding channel
u32MaxWidth	Maximum width of screenshot, default 4096
u32MaxHeight	Maximum height of screenshot, default 4096
pfnDataCallback	Screenshot data callback pointer

Related data types and interfaces

RKADK_PPLAYER_SNAPSHOT_RECV_FN

RKADK_PLAYER_CFG_S

6.3.12 RKADK_PLAYER_VDEC_CFG_S

Description

Define the VDEC attribute structure.

Definition

```
typedef struct {
  RKADK_U32 u32FrameBufCnt; //frame buffer cnt(output), default: 3
  RKADK_U32 u32StreamBufCnt; //stream buffer cnt(input), default: 3
} RKADK_PLAYER_VDEC_CFG_S;
```

Members

Member Name	Description
u32FrameBufCnt	Number of output buffers, it is 3 by default
u32StreamBufCnt	Number of input buffers, it is 3 by default

Related data types and interfaces

RKADK_PLAYER_CFG_S

6.3.13 RKADK_PLAYER_RTSP_CFG_S

Description

Define the RTSP attribute structure.

Definition

```
typedef struct {
  const char *transport; //udp or tcp, default: udp
  RKADK_U32 u32IoTimeout; //timeout (in microseconds) of socket I/O operations
} RKADK_PLAYER_RTSP_CFG_S;
```

Members

Member Name	Description
transport	Transport protocol, it is UDP by default
u32ITimeout	Socket I/O operation timeout duration, unit is ms, it is no timeout by default

Related data types and interfaces

RKADK_PLAYER_CFG_S

6.3.14 RKADK_PLAYER_PACKET

Description

Define the data packet structure and enable the use of the third-party demuxer library.

Definition

```
typedef struct {
  bool bEofFlag;
  RKADK_S8 *s8PacketData;
  RKADK_S32 s32PacketSize;
  RKADK_U32 u32Seq;
  RKADK_S64 s64Pts;

  //if bypass, must set pFreeCB;
  bool bBypass;
  RKADK_MPI_MB_FREE_CB pFreeCB;
} RKADK_PLAYER_PACKET;
```

Members

Member Name	Description
bEofFlag	Whether it is the last frame of data
s8PacketData	Data pointer
s32PacketSize	Data length
u32Seq	Serial number
s64Pts	Timestamp
bBypass	Data transmission method, true: pass-through, false: secondary copy
pFreeCB	The s8PacketData releases the function pointer, it must be set when bBypass is true

Related data types and interfaces

RKADK_PLAYER_SendAudioPacket

RKADK_PLAYER_SendVideoPacket

6.3.15 RKADK_PLAYER_DATA_PARAM_S

Description

Define the audio and video parameter structure, it can be used when enabling the third-party demuxer library.

Definition

```
typedef struct {
  const RKADK_CHAR *pFilePath;
  RKADK_BOOL bIsRtsp;
  RKADK_BOOL bVideoExist;
  RKADK_BOOL bAudioExist;

  //video param
  RKADK_CODEC_TYPE_E enVCodecType;
  RKADK_U32 u32Width;
  RKADK_U32 u32Height;
  RKADK_FORMAT_E enPixFmt; //output pixel format
  RKADK_U32 u32FrameRate;

  //audio param
  RKADK_CODEC_TYPE_E enACodecType;
  RKADK_S32 u32BitWidth;
  RKADK_S32 u32SampleRate;
  RKADK_S32 u32Channel;
} RKADK_PLAYER_DATA_PARAM_S;
```

Members

Member Name	Description
pFilePath	The path of the file to be played
bIsRtsp	Whether it is RTSP network stream
bVideoExist	Whether there is a video stream in the file to be played
bAudioExist	Whether there is an audio stream in the file to be played
enVCodecType	Video stream decoding format
u32Width	Video stream width
u32Height	Video stream high
u32FrameRate	Video stream frame rate
enACodecType	Audio stream decoding format
u32BitWidth	Audio stream bit width
u32SampleRate	Audio stream sample rate
u32Channel	Number of audio stream channels

Related data types and interfaces

RKADK_PLAYER_SetDataParam

7. Live Streaming

7.1 Overview

Provides basic services of standard RTSP live streaming; provides RTMP live streaming services. RTSP and RTMP do not support starting at the same time.

7.2 API Reference

7.2.1 RTSP

7.2.1.1 RKADK_RTSP_Init

Description

Initialize the RTSP module.

Syntax

RKADK_S32 RKADK_RTSP_Init(RKADK_U32 u32CamId, RKADK_U32 port, const char *path, RKADK_MW_PTR *ppHandle);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
port	Port number	Input
path	RTSP address	Input
ppHandle	Created RTSP Handle	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_rtsp.h

Library file: librkadk.so

Notice

- Call RKADK_RTSP_Start after RKADK_RTSP_Init to start RTSP live streaming.

Example

rkadk_rtsp_test

See Also

RKADK_RTSP_DeInit

7.2.1.2 RKADK_RTSP_DeInit

Description

Deinitialize the RTSP module.

Syntax

RKADK_S32 RKADK_RTSP_DeInit(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Created RTSP Handle	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_rtsp.h

Library file: librkadk.so

Example

rkadk_rtsp_test

See Also

RKADK_RTSP_Init

7.2.1.3 RKADK_RTSP_Start

Description

Start RTSP live streaming.

Syntax

RKADK_S32 RKADK_RTSP_Start(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Created RTSP Handle	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_rtsp.h

Library file: librkadk.so

Notice

- Call this interface after RKADK_RTSP_Init.

Example

rkadk_rtsp_test

See Also

RKADK_RTSP_Stop

7.2.1.4 RKADK_RTSP_Stop

Description

Stop RTSP live streaming.

Syntax

RKADK_S32 RKADK_RTSP_Stop(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Created RTSP Handle	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_rtsp.h

Library file: librkadk.so

Example

rkadk_rtsp_test

See Also

RKADK_RTSP_Start

7.2.1.5 RKADK_RTSP_VideoReset

Description

Reconfigure the RTSP live streaming.

Syntax

RKADK_S32 RKADK_RTSP_VideoReset(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Created RTSP Handle	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_rtsp.h

Library file: librkadk.so

Notice

- The RKADK_RTSP_VideoReset interface can only be used after RKADK_RTSP_Init.
- When switching resolution, frame rate, bit rate, encoding type, or recording type, reset the parameters of the Live module in the ini file by using the RKADK_PARAM_SetCamParam API.
- RV1126/RV1109 and RK3576 does not support dynamic switching of resolution and encoding type. You need to RKADK_RTSP_DeInit first, then configure the new resolution or encoding type, and finally RKADK_RTSP_Init again.
- If Live and other modules share the VI channel, both the resolution of Live and multiplexing modules need to be switched simultaneously to avoid VI and VENC resolution mismatch.

7.2.2 RTMP

7.2.2.1 RKADK_RTMP_Init

Description

Initialize RTMP module.

Syntax

RKADK_S32 RKADK_RTMP_Init(RKADK_U32 u32CamId, const char *path, RKADK_MW_PTR *ppHandle);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
path	RTMP address	Input
ppHandle	Created RTMP Handle	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_rtmp.h

Library file: librkadk.so

Example

rkadk_rtmp_test

See Also

RKADK_RTMP_DeInit

7.2.2.2 RKADK_RTMP_DeInit

Description

Deinitialize the RTMP module.

Syntax

RKADK_S32 RKADK_RTMP_DeInit(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Created RTMP Handle	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_rtmp.h

Library file: librkadk.so

Example

rkadk_rtmp_test

See Also

RKADK_RTMP_Init

7.2.2.3 RKADK_RTSP_VideoReset

Description

Reconfigure the RTSP live streaming.

Syntax

RKADK_S32 RKADK_RTSP_VideoReset(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Created RTSP Handle	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_rtsp.h

Library file: librkadk.so

Notice

- The RKADK_RTMP_VideoReset interface can only be used after RKADK_RTMP_Init.
- When switching resolution, frame rate, bit rate, encoding type, or recording type, reset the parameters of the Live module in the ini file by using the RKADK_PARAM_SetCamParam API.
- RV1126/RV1109 and RK3576 does not support dynamic switching of resolution and encoding type. You need to RKADK_RTMP_DeInit first, then configure the new resolution or encoding type, and finally RKADK_RTMP_Init again.
- If Live and other modules share the VI channel, both the resolution of Live and multiplexing modules need to be switched simultaneously to avoid VI and VENC resolution mismatch.

8. Storage

8.1 Overview

Provides basic storage functions, currently including the following functions:

- File detection, storage, acquisition, management
- Device capacity and status query
- Automatically delete files
- Format

8.2 API Reference

8.2.1 RKADK_STORAGE_Init

Description

Storage module initialization.

Syntax

RKADK_S32 RKADK_STORAGE_Init(RKADK_MW_PTR *ppHandle, RKADK_STR_DEV_ATTR *pstDevAttr);

Parameters

Parameter name	Description	Input/output
ppHandle	Created storage module handle	Output
pstDevAttr	Mounted device attribute pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_storage.h

Library file: librkadk.so

Notice

- Repeated initialization is not supported.
- If RKADK_STR_DEV_ATTR pass in NULL, default attributes are used. The default attributes are: 2 folders, named video_front and video_back, sorted by file name, automatic deletion threshold 500~1000M, limit the number of files not used, with the ratio of 50%.

Example

rkadk_storage_test

See Also

RKADK_STORAGE_Deinit

8.2.2 RKADK_STORAGE_Deinit

Description

Storage module deinitialization.

Syntax

RKADK_S32 RKADK_STORAGE_Deinit(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Storage module handle pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_storage.h

Library file: librkadk.so

Notice

- Repeated deinitialization is not supported.

Example

rkadk_storage_test

See Also

RKADK_STORAGE_Init

8.2.3 RKADK_STORAGE_GetDevAttr

Description

Get the mounted device attribute.

Syntax

RKADK_STR_DEV_ATTR RKADK_STORAGE_GetDevAttr(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Storage module handle pointer	Input

Return value

Return value	Description
RKADK_STR_DEV_ATTR	Mounted device attribute structure

Requirement

Header file: rkadk_storage.h

Library file: librkadk.so

Notice

- The RKADK_STORAGE_GetDevAttr interface can only be used after the storage module is initialized.

Example

None

See Also

None

8.2.4 RKADK_STORAGE_GetMountStatus

Description

Get device mounting status.

Syntax

RKADK_MOUNT_STATUS RKADK_STORAGE_GetMountStatus(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Storage module handle pointer	Input

Return value

Return value	Description
RKADK_MOUNT_STATUS	Mount status enumeration type

Requirement

Header file: rkadk_storage.h

Library file: librkadk.so

Example

rkadk_storage_test

See Also

None

8.2.5 RKADK_STORAGE_GetCapacity

Description

Get the device capacity.

Syntax

RKADK_S32 RKADK_STORAGE_GetCapacity(RKADK_MW_PTR *ppHandle, RKADK_S32 *totalSize, RKADK_S32 *freeSize);

Parameters

Parameter name	Description	Input/output
ppHandle	Created storage module handle	Input/Output
totalSize	Total device capacity pointer	Output
freeSize	Device remaining capacity pointer	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_storage.h

Library file: librkadk.so

Example

rkadk_storage_test

See Also

None

8.2.6 RKADK_STORAGE_GetFileList

Description

Get a list of files.

Syntax

RKADK_S32 RKADK_STORAGE_GetFileList(RKADK_FILE_LIST *list, RKADK_MW_PTR pHandle, RKADK_SORT_TYPE sort);

Parameters

Parameter name	Description	Input/output
list	File list structure pointer	Input/output
PHandle	Storage module handle pointer	Input
sort	Sort type	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_storage.h

Library file: librkadk.so

Notice

- Must be used together with RKADK_STORAGE_FreeFileList.

Example

rkadk_storage_test

See Also

RKADK_STORAGE_FreeFileList

8.2.7 RKADK_STORAGE_FreeFileList

Description

Release file list.

Syntax

RKADK_S32 RKADK_STORAGE_FreeFileList(RKADK_FILE_LIST *list);

Parameters

Parameter name	Description	Input/output
list	file list structure pointer	Input/output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_storage.h

Library file: librkadk.so

Notice

- Must be used together with RKADK_STORAGE_GetFileList.

Example

rkadk_storage_test

See Also

RKADK_STORAGE_GetFileList

8.2.8 RKADK_STORAGE_GetFileNum

Description

Get the number of files.

Syntax

RKADK_S32 RKADK_STORAGE_GetFileNum(RKADK_CHAR *fileListPath, RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
fileListPath	File list path pointer	Input
PHandle	Storage module handle pointer	Input

Return value

Return value	Description
non-negative	Number of files
-1	Failure

Requirement

Header file: rkadk_storage.h

Library file: librkadk.so

Example

None

See Also

None

8.2.9 RKADK_STORAGE_GetDevPath

Description

Get the path of the mounted device.

Syntax

RKADK_CHAR *RKADK_STORAGE_GetDevPath(RKADK_MW_PTR pHandle);

Parameters

Parameter name	Description	Input/output
PHandle	Storage module handle pointer	Input

Return value

Return value	Description
RKADK_CHAR *	Mount device path pointer

Requirement

Header file: rkadk_storage.h

Library file: librkadk.so

Notice

- The RKADK_STORAGE_GetDevPath interface can only be used after the storage module is initialized.

Example

rkadk_storage_test

See Also

None

8.2.10 RKADK_STORAGE_Format

Description

Device formatting.

Syntax

RKADK_S32 RKADK_STORAGE_Format(RKADK_MW_PTR pHandle, RKADK_CHAR *cFormat);

Parameters

Parameter name	Description	Input/output
PHandle	Storage module handle pointer	Input
cFormat	File system type	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_storage.h

Library file: librkadk.so

Example

None

See Also

None

8.3 Type of Data

The storage module mainly provides the following data types:

RKADK_MOUNT_STATUS: Mount status enumeration type

RKADK_SORT_TYPE: Sort type enumeration

RKADK_SORT_CONDITION: Sort condition enumeration type

RKADK_STR_FOLDER_ATTR: Folder attribute structure

RKADK_STR_DEV_ATTR: Device attribute structure

RKADK_FILE_INFO: File information structure

RKADK_FILE_LIST: File list structure

RKADK_FILE_LIST_ARRAY: File list group structure

8.3.1 RKADK_MOUNT_STATUS

Description

Defines the mount status enumeration type.

Definition

```
typedef enum {
  DISK_UNMOUNTED = 0,
  DISK_NOT_FORMATTED,
  DISK_FORMAT_ERR,
  DISK_SCANNING,
  DISK_MOUNTED,
  DISK_MOUNT_BUTT,
} RKADK_MOUNT_STATUS;
```

Members

Member Name	Description
DISK_UNMOUNTED	The disk is not mounted
DISK_NOT_FORMATTED	The disk is not formatted
DISK_FORMAT_ERR	Disk format error
DISK_SCANNING	The disk is being scanned
DISK_MOUNTED	The disk is mounted

Related data types and interfaces

RKADK_STORAGE_GetMountStatus

8.3.2 RKADK_SORT_TYPE

Description

Defines the sorting type enum.

Definition

```
typedef enum {
  LIST_ASCENDING = 0,
  LIST_DESCENDING,
  LIST_BUTT,
} RKADK_SORT_TYPE;
```

Members

Member Name	Description
LIST_ASCENDING	Sort the list in ascending order
LIST_DESCENDING	Sort the list in descending order

Related data types and interfaces

RKADK_STORAGE_GetFileList

8.3.3 RKADK_SORT_CONDITION

Description

Defines the sorting condition enumeration type.

Definition

```
typedef enum {
  SORT_MODIFY_TIME = 0,
  SORT_FILE_NAME,
  SORT_BUTT,
} RKADK_SORT_CONDITION;
```

Members

Member Name	Description
SORT_MODIFY_TIME	The list is sorted by file modification time
SORT_FILE_NAME	List sorted by file name

Related data types and interfaces

RKADK_STR_FOLDER_ATTR

8.3.4 RKADK_STR_FOLDER_ATTR

Description

Define the folder attribute structure.

Definition

```
typedef struct {
  RKADK_CHAR cFolderPath[RKADK_MAX_FILE_PATH_LEN];
  RKADK_SORT_CONDITION s32SortCond;
  RKADK_BOOL bNumLimit;
  RKADK_S32 s32Limit;
} RKADK_STR_FOLDER_ATTR;
```

Members

Member Name	Description
cFolderPath	Folder path
s32SortCond	Sorting conditions
bNumLimit	Option: Whether to set the upper limit based on the number of files
s32Limit	Maximum folder capacity (ratio/number)

Related data types and interfaces

RKADK_SORT_CONDITION

RKADK_STR_DEV_ATTR

8.3.5 RKADK_STR_DEV_ATTR

Description

Define the device attribute structure.

Definition

```
typedef struct {
  RKADK_CHAR cDevPath[RKADK_MAX_FILE_PATH_LEN];
  RKADK_CHAR cMountPath[RKADK_MAX_FILE_PATH_LEN];
  RKADK_S32 s32FreeSizeDelMin;
  RKADK_S32 s32FreeSizeDelMax;
  RKADK_S32 s32AutoDel;
  RKADK_S32 s32FolderNum;
  RKADK_CHAR cFormatId[RKADK_MAX_FORMAT_ID_LEN];
  RKADK_CHAR cVolume[RKADK_MAX_VOLUME_LEN];
  RKADK_S32 s32CheckFormatId;
  RKADK_STR_FOLDER_ATTR *pstFolderAttr;
} RKADK_STR_DEV_ATTR;
```

Members

Member Name	Description
cDevPath	Device name (device path)
cMountPath	Device mount path
s32FreeSizeDelMin	Automatically delete the lower limit of the threshold
s32FreeSizeDelMax	Automatically delete the upper limit of the threshold
s32AutoDel	Automatic deletion option
s32FolderNum	Number of folders
cFormatId	Format ID
cVolume	Volume label
s32CheckFormatId	Check whether the format ID matches
pstFolderAttr	Folder attribute structure pointer

Related data types and interfaces

RKADK_STR_FOLDER_ATTR

RKADK_STORAGE_Init

RKADK_STORAGE_GetDevAttr

8.3.6 RKADK_FILE_INFO

Description

Define the file information structure.

Definition

```
typedef struct {
  RKADK_CHAR filename[RKADK_MAX_FILE_PATH_LEN];
  off_t stSize;
  time_t stTime;
  void *thumb;
} RKADK_FILE_INFO;
```

Members

Member Name	Description
filename	File name
stSize	File size
stTime	File modification time
thumb	Thumbnail pointer

Related data types and interfaces

RKADK_FILE_LIST

8.3.7 RKADK_FILE_LIST

Description

Define the file list structure.

Definition

```
typedef struct {
  RKADK_CHAR path[RKADK_MAX_FILE_PATH_LEN];
  RKADK_S32 s32FileNum;
  RKADK_FILE_INFO *file;
} RKADK_FILE_LIST;
```

Members

Member Name	Description
path	File list (folder) path
s32FileNum	Number of files
file	File information structure pointer

Related data types and interfaces

RKADK_FILE_INFO

RKADK_FILE_LIST_ARRAY

8.3.8 RKADK_FILE_LIST_ARRAY

Description

Define the file list group structure.

Definition

```
typedef struct {
  RKADK_S32 s32ListNum;
  RKADK_FILE_LIST *list;
} RKADK_FILE_LIST_ARRAY;
```

Members

Member Name	Description
s32ListNum	Number of file lists (folders)
list	File list structure pointer

Related data types and interfaces

RKADK_FILE_LIST

9. Local preview

9.1 Overview

Provides local preview function.

9.2 API Reference

9.2.1 RKADK_DISP_Init

Description

Initialize the local preview module.

Syntax

RKADK_S32 RKADK_DISP_Init(RKADK_U32 u32CamId);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_disp.h

Library file: librkadk.so

Example

rkadk_disp_test

See Also

RKADK_DISP_DeInit

9.2.2 RKADK_DISP_DeInit

Description

Deinitialize the local preview module.

Syntax

RKADK_S32 RKADK_DISP_DeInit(RKADK_U32 u32CamId);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_disp.h

Library file: librkadk.so

Example

rkadk_disp_test

See Also

RKADK_DISP_Init

9.2.3 RKADK_DISP_SetAttr

Description

Set preview properties.

Syntax

RKADK_S32 RKADK_DISP_SetAttr(RKADK_U32 u32CamId, RKADK_DISP_ATTR_S *pstAttr);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
pstAttr	preview attributes	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_disp.h

Library file: librkadk.so

9.3 Type of Data

9.3.1 RKADK_DISP_ATTR_S

Description

Define the preview attribute structure.

Definition

```
typedef struct {
  RKADK_RECT_S stVpssCropRect;
  RKADK_RECT_S stVoRect;
} RKADK_DISP_ATTR_S;
```

Members

Member Name	Description
stVpssCropRect	Input display area
stVoRect	Output display area

10. Watermark

10.1 Overview

The Watermark module provides basic watermark functionality

10.2 API Reference

10.2.1 RKADK_OSD_Init

Description

Initialize the watermark task.

Syntax

RKADK_S32 RKADK_OSD_Init(RKADK_U32 u32OsdId, RKADK_OSD_ATTR_S *pstOsdAttr);

Parameters

Parameter name	Description	Input/output
u32OsdId	Watermark ID	Input
pstOsdAttr	Watermark task attribute pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_osd.h

Library file: librkadk.so

Notice

- Repeated initialization is not supported.

10.2.2 RKADK_OSD_Deinit

Description

Deinitialize the watermark task.

Syntax

RKADK_S32 RKADK_OSD_Deinit(RKADK_U32 u32OsdId);

Parameters

Parameter name	Description	Input/output
u32OsdId	Watermark ID	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_osd.h

Library file: librkadk.so

Notice

- Repeated deinitialization is not supported.

10.2.3 RKADK_OSD_UpdateBitMap

Description

Watermark content updated.

Syntax

RKADK_S32 RKADK_OSD_UpdateBitMap(RKADK_U32 u32OsdId, RKADK_OSD_ATTR_S *pstOsdAttr);

Parameters

Parameter name	Description	Input/output
u32OsdId	Watermark ID	Input
pstOsdAttr	Watermark task attribute pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_osd.h

Library file: librkadk.so

Notice

- Must be used after the de-initialization task is completed.

10.2.4 RKADK_OSD_AttachToStream

Description

Attach watermark to the target stream.

Syntax

RKADK_S32 RKADK_OSD_AttachToStream(RKADK_U32 u32OsdId, RKADK_U32 u32CamId, RKADK_STREAM_TYPE_E enStrmType, RKADK_OSD_STREAM_ATTR_S *pstOsdStreamAttr);

Parameters

Parameter name	Description	Input/output
u32OsdId	Watermark ID	Input
u32CamId	Camera ID	Input
enStrmType	Target stream type	Input
pstOsdStreamAttr	Watermark position information pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_osd.h

Library file: librkadk.so

Notice

- The same watermark cannot be attached repeatedly on the same stream type.

10.2.5 RKADK_OSD_DettachFromStream

Description

Remove watermark from the target stream.

Syntax

RKADK_S32 RKADK_OSD_DettachFromStream(RKADK_U32 u32OsdId, RKADK_U32 u32CamId, RKADK_STREAM_TYPE_E enStrmType);

Parameters

Parameter name	Description	Input/output
u32OsdId	Watermark ID	Input
u32CamId	Camera ID	Input
enStrmType	Target stream type	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_osd.h

Library file: librkadk.so

10.2.6 RKADK_OSD_UpdateOsdSize

Description

Update watermark size.

Syntax

RKADK_S32 RKADK_OSD_UpdateOsdSize(RKADK_U32 u32OsdId, RKADK_OSD_ATTR_S *pstOsdAttr);

Parameters

Parameter name	Description	Input/output
u32OsdId	Watermark ID	Input
pstOsdAttr	Watermark task attribute pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_osd.h

Library file: librkadk.so

10.2.7 RKADK_OSD_UpdateDisplayAttr

Description

Update the watermark display area.

Syntax

RKADK_S32 RKADK_OSD_UpdateDisplayAttr(RKADK_U32 u32OsdId, RKADK_U32 u32CamId, RKADK_STREAM_TYPE_E enStrmType, RKADK_OSD_STREAM_ATTR_S *pstOsdStreamAttr);

Parameters

Parameter name	Description	Input/output
u32OsdId	Watermark ID	Input
u32CamId	Camera ID	Input
enStrmType	Stream type	Input
pstOsdStreamAttr	Watermark location information	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_osd.h

Library file: librkadk.so

10.3 Type of Data

The watermark module mainly provides the following data types:

RKADK_OSD_ATTR_S: Watermark attribute structure
RKADK_OSD_STREAM_ATTR_S: Watermark location information structure

RKADK_OSD_TYPE_E: Watermark overlay type

10.3.1 RKADK_OSD_ATTR_S

Description

Define watermark attributes.

Definition

```
typedef struct {
  RKADK_U32 Width;
  RKADK_U32 Height;
  RKADK_VOID *pData;
  RKADK_FORMAT_E Format;
  RKADK_OSD_TYPE_E enOsdType;
} RKADK_OSD_ATTR_S;
```

Members

Member Name	Description
Width	Watermark width
Height	Watermark height
pData	Watermark content
Format	Watermark format
enOsdType	Watermark overlay type

Related data types and interfaces

RKADK_OSD_ATTR_S

RKADK_OSD_TYPE_E

10.3.2 RKADK_OSD_STREAM_ATTR_S

Description

Define watermark location information.

Definition

```
typedef struct {
  RKADK_BOOL bEnableShow;
  RKADK_U32 Origin_X;
  RKADK_U32 Origin_Y;
} RKADK_OSD_STREAM_ATTR_S;
```

Members

Member Name	Description
Origin_X	Watermark starting position X offset
Origin_Y	Watermark starting position Y offset
bEnableShow	Whether to display watermark

Related data types and interfaces

RKADK_OSD_STREAM_ATTR_S

10.4 RKADK_OSD_TYPE_E

Description

Define the watermark overlay type.

Definition

```
typedef enum {
  RKADK_OSD_TYPE_NORMAL = 0, //use encoder do osd
  RKADK_OSD_TYPE_EXTRA,      //use rga do osd
  RKADK_OSD_TYPE_BUTT
} RKADK_OSD_TYPE_E;
```

Members

Member Name	Description
RKADK_OSD_TYPE_NORMAL	Encoder OSD overlay
RKADK_OSD_TYPE_EXTRA	RGA overlay

Notice

- RV1109/RV1126 JPEG encoding does not support the encoder OSD overlay watermark, and needs to be configured to RKADK_OSD_TYPE_EXTRA.

11. UI Overlay

11.1 Overview

Provides UI overlay function for overlaying UI and video images on single-layer VOP platforms such as RV1103/RV1106.

11.2 API Reference

11.2.1 RKADK_UI_Create

Description

Initialize the UI overlay module

Syntax

RKADK_S32 RKADK_UI_Create(RKADK_UI_ATTR_S *pstUiAttr, RKADK_MW_PTR *ppUi);

Parameters

Parameter name	Description	Input/output
pstUiAttr	UI attributes	Input
ppUi	Created UI task pointer	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_ui.h

Library file: librkadk.so

Example

rkadk_ui_test

See Also

RKADK_UI_Destroy

11.2.2 RKADK_UI_Destroy

Description

Deinitialize UI overlay module

Syntax

RKADK_S32 RKADK_UI_Destroy(RKADK_MW_PTR pUi);

Parameters

Parameter name	Description	Input/output
pUi	UI task pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_ui.h

Library file: librkadk.so

Example

rkadk_ui_test

See Also

RKADK_UI_Create

11.2.3 RKADK_UI_Update

Description

Refresh UI data.

Syntax

RKADK_S32 RKADK_UI_Update(RKADK_MW_PTR pUi, RKADK_UI_FRAME_INFO *pstUiFrameInfo);

Parameters

Parameter name	Description	Input/output
pUi	UI task pointer	Input
pstUiFrameInfo	UI data pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_ui.h

Library file: librkadk.so

Example

rkadk_ui_test

11.3 Type of Data

RKADK_UI_ATTR_S: UI attribute structure

RKADK_UI_FRAME_INFO: UI data structure

11.3.1 RKADK_UI_ATTR_S

Description

Define the UI attribute structure.

Definition

```
typedef struct {
  RKADK_U32 u32DispX;
  RKADK_U32 u32DispY;
  RKADK_U32 u32DispWidth;
  RKADK_U32 u32DispHeight;
  RKADK_U32 u32DispFrmRt;
  RKADK_U32 u32ImgWidth;
  RKADK_U32 u32ImgHeight;
  RKADK_U32 u32VoLay;
  RKADK_U32 u32VoDev;
  RKADK_U32 u32VoChn;
  RKADK_U32 u32Rotation; //0: 0, 1: 90, 2: 180, 3: 270
  RKADK_BOOL bMirror;
  RKADK_BOOL bFlip;
  RKADK_VO_FORMAT_E enUiVoFormat;
  RKADK_VO_INTF_TYPE_E enUiVoIntfTye;
  RKADK_VO_SPLICE_MODE_E enVoSpliceMode;
} RKADK_UI_ATTR_S;
```

Members

Member Name	Description
u32DispX	Layer display area x coordinate
u32DispY	Layer display area y coordinate
u32DispWidth	Layer display area width
u32DispHeight	Layer display area height
u32ImgWidth	Layer image width
u32ImgHeight	Layer image height
u32DispFrmRt	Refresh frame rate
u32VoLay	Video output video layer number
u32VoDev	Display output device number
u32VoChn	Video output channel number, value range: [0, VO_MAX_CHN_NUM(128)]
u32Rotation	Rotation, values: [0: 0, 1: 90, 2: 180, 3: 270]
bMirror	Enable mirror
bFlip	Enable flip
enUiVoFormat	Image pixel format
enUiVoIntfTye	Display interface type
enVoSpliceMode	Layer composition mode

Related data types and interfaces

RKADK_VO_FORMAT_E

RKADK_VO_INTF_TYPE_E

RKADK_VO_SPLICE_MODE_E

RKADK_UI_Create

11.3.2 RKADK_UI_FRAME_INFO

Description

Define UI data information.

Definition

```
typedef struct {
  RKADK_U32 u32Width;
  RKADK_U32 u32Height;
  RKADK_FORMAT_E Format;
  RKADK_VOID *pMblk;
  RKADK_S32 s32MilliSec; //update timeout in millisecond, default: 1000ms
} RKADK_UI_FRAME_INFO;
```

Members

Member Name	Description
u32Width	UI data width
u32Height	UI data height
Format	UI data format
pMblk	UI data pointer
s32MilliSec	UI update timeout, default 1000ms, -1 is block

Related data types and interfaces

RKADK_FORMAT_E

RKADK_UI_Update

11.3.3 RKADK_FORMAT_E

Description

Define the pixel format.

Definition

```
typedef enum {
  RKADK_FMT_ARGB1555,                                /* 16-bit RGB
  */
  RKADK_FMT_ABGR1555,                                /* 16-bit RGB
  */
  RKADK_FMT_RGBA5551,                                /* 16-bit RGB
  */
  RKADK_FMT_BGRA5551,                                /* 16-bit RGB
  */
  RKADK_FMT_ARGB4444,                                /* 16-bit RGB
  */
  RKADK_FMT_ABGR4444,                                /* 16-bit RGB
  */
  RKADK_FMT_RGBA4444,                                /* 16-bit RGB
  */
  RKADK_FMT_BGRA4444,                                /* 16-bit RGB
  */
  RKADK_FMT_ARGB8888,                                /* 32-bit RGB
  */
```

```
  RKADK_FMT_ABGR8888,                                 /* 32-bit RGB
  */
  RKADK_FMT_RGBA8888,                                 /* 32-bit RGB
  */
  RKADK_FMT_BGRA8888,                                 /* 32-bit RGB
  */
  RKADK_FMT_2BPP,
  RKADK_FMT_YUV420SP,
  RKADK_FMT_YUV420SP_10BIT,
  RKADK_FMT_YUV422SP,
  RKADK_FMT_BUTT,
} RKADK_FORMAT_E;
```

Related data types and interfaces

RKADK_UI_FRAME_INFO

12. Parameter Settings

12.1 Overview

The parameter setting module is strongly related to the product form. By combining and using common component data structures, a data structure suitable for the product form is defined.

This module supports obtaining specified parameters, saving specified parameters, and restoring parameters to default.

To facilitate editing, parameters are stored in the form of ini files.

You can specify the default ini path by setting the environment variable rkadk_default_ini_path. The default path is /oem/usr/etc.

```
export rkadk_default_ini_path=/oem/usr/etc
```

12.2 API Reference

12.2.1 RKADK_PARAM_Init

Description

Initialization parameter module

Syntax

RKADK_S32 RKADK_PARAM_Init(char *globalSetting, char **sesnorSettingArrary);

Parameters

Parameter name	Description	Input/output
globalSetting	Global ini configuration file path	Input
sesnorSettingArrary	Sensor ini configuration file path	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_param.h

Library file: librkadk.so

Notice

- Before starting any module, RKADK_PARAM_Init should be called firstly to initialize the parameter module.
- If globalSetting is not set, the default path RKADK_PARAM_PATH is used.
- If sesnorSettingArrary is not set, the default path RKADK_PARAM_PATH_SENSOR_PREFIX is used. The sensor configuration file prefix defaults to rkadk_setting_sensor_n.ini, _n is the sensor camera ID, and the serial number starts from 0.

Example

rkadk_record_test

See Also

None

12.2.2 RKADK_PARAM_GetCamParam

Description

Get Camera related parameters.

Syntax

RKADK_S32 RKADK_PARAM_GetCamParam(RKADK_S32 s32CamID, RKADK_PARAM_TYPE_E enParamType, RKADK_VOID *pvParam);

Parameters

Parameter name	Description	Input/output
s32CamID	Camera ID	Input
enParamType	Parameter type	Input
pvParam	Obtained parameter pointer	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_param.h

Library file: librkadk.so

Notice

- This interface can be called only after calling RKADK_PARAM_Init to initialize the parameter module.

Example

rkadk_record_test

See Also

RKADK_PARAM_SetCamParam

12.2.3 RKADK_PARAM_SetCamParam

Description

Set Camera related parameters.

Syntax

RKADK_S32 RKADK_PARAM_SetCamParam(RKADK_S32 s32CamID, RKADK_PARAM_TYPE_E enParamType, const RKADK_VOID *pvParam);

Parameters

Parameter name	Description	Input/output
s32CamID	Camera ID	Input
enParamType	Parameter type	Input
pvParam	Set parameter pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_param.h

Library file: librkadk.so

Notice

- This interface can be called only after calling RKADK_PARAM_Init to initialize the parameter module.

Example

rkadk_record_test

See Also

RKADK_PARAM_GetCamParam

12.2.4 RKADK_PARAM_GetCommParam

Description

Get common parameters.

Syntax

RKADK_S32 RKADK_PARAM_GetCommParam(RKADK_PARAM_TYPE_E enParamType, RKADK_VOID *pvParam);

Parameters

Parameter name	Description	Input/output
enParamType	Parameter type	Input
pvParam	Obtained parameter pointer	Output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_param.h

Library file: librkadk.so

Notice

- This interface can be called only after calling RKADK_PARAM_Init to initialize the parameter module.

Example

rkadk_record_test

See Also

RKADK_PARAM_SetCommParam

12.2.5 RKADK_PARAM_SetCommParam

Description

Set common parameters.

Syntax

RKADK_S32 RKADK_PARAM_SetCommParam(RKADK_PARAM_TYPE_E enParamType, const RKADK_VOID *pvParam);

Parameters

Parameter name	Description	Input/output
enParamType	Parameter type	Input
pvParam	Set parameter pointer	Input

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_param.h

Library file: librkadk.so

Notice

- This interface can be called only after calling RKADK_PARAM_Init to initialize the parameter module.

Example

rkadk_record_test

See Also

RKADK_PARAM_GetCommParam

12.2.6 RKADK_PARAM_SetDefault

Description

Restore default configuration.

Syntax

RKADK_S32 RKADK_PARAM_SetDefault(RKADK_VOID);

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_param.h

Library file: librkadk.so

12.2.7 RKADK_PARAM_GetResolution

Description

RKADK_PARAM_RES_E is used to convert to specific resolution.

Syntax

RKADK_S32 RKADK_PARAM_GetResolution(RKADK_PARAM_RES_E type, RKADK_U32 *width, RKADK_U32 *height);

Parameters

Parameter name	Description	Input/output
type	Resolution type	Input
width	Converted resolution width	output
height	Converted resolution high	output

Return value

Return value	Description
0	Success
Other value	Failure

Requirement

Header file: rkadk_param.h

Library file: librkadk.so

Notice

- This interface can be called only after calling RKADK_PARAM_Init to initialize the parameter module.

Example

None

See Also

RKADK_PARAM_GetResType

12.2.8 RKADK_PARAM_GetResType

Description

Convert resolution to RKADK_PARAM_RES_E.

Syntax

RKADK_PARAM_RES_E RKADK_PARAM_GetResType(RKADK_U32 width, RKADK_U32 height);

Parameters

Parameter name	Description	Input/output
width	Resolution width	Input
height	Resolution height	Input

Return value

Return value	Description
Corresponding RKADK_PARAM_RES_E	Success
RKADK_RES_BUTT	Failure

Requirement

Header file: rkadk_param.h

Library file: librkadk.so

Notice

- This interface can be called only after calling RKADK_PARAM_Init to initialize the parameter module.

Example

None

See Also

RKADK_PARAM_GetResolution

12.2.9 RKADK_PARAM_GetVencChnId

Description

Get the VENC channel number corresponding to Record, Photo, and Stream.

Syntax

RKADK_S32 RKADK_PARAM_GetVencChnId(RKADK_U32 u32CamId, RKADK_STREAM_TYPE_E enStrmType);

Parameters

Parameter name	Description	Input/output
u32CamId	Camera ID	Input
enStrmType	Stream type	Input

Return value

Return value	Description
Corresponding VENC channel number	Success
-1	Failure

Requirement

Header file: rkadk_param.h

Library file: librkadk.so

Notice

- This interface can be called only after calling RKADK_PARAM_Init to initialize the parameter module.

Example

None

See Also

None

12.3 Type of Data

The parameter module mainly provides the following data types:

RKADK_DEFPARAM_PATH: Default global ini configuration file path

RKADK_DEFPARAM_PATH_SENSOR_PREFIX: Default Sensor ini configuration file path

RKADK_PARAM_PATH: Global ini configuration file path

RKADK_PARAM_PATH_SENSOR_PREFIX: Sensor ini configuration file path

RKADK_PARAM_TYPE_E: Parameter type enumeration

RKADK_PARAM_RES_E: Resolution type enumeration

RKADK_STREAM_TYPE_E: Data stream type enumeration

RKADK_PARAM_CODEC_CFG_S: Coding type configuration structure

RKADK_PARAM_BITRATE_S: Bit rate configuration structure

RKADK_PARAM_REC_TIME_S: Recording duration configuration structure

RKADK_PARAM_GOP_S: VENC GOP configuration structure

RKADK_VQE_MODE_E: Audio input sound quality enhancement enumeration

RKADK_MUXER_FILE_TYPE_E: Recording file type enumeration

RKADK_MUXER_PRE_RECORD_MODE_E: Pre-record mode enumeration

RKADK_MIC_TYPE_E: Audio device channel mode type enumeration

12.3.1 RKADK_DEFPARAM_PATH

Description

The default global ini configuration file path stores the configuration shared by each sensor and is used to restore the default configuration.

Definition

```
#define RKADK_DEFPARAM_PATH "/oem/usr/etc/rkadk_defsetting.ini"
```

12.3.2 RKADK_DEFPARAM_PATH_SENSOR_PREFIX

Description

The default sensor ini configuration file path stores the unique configuration of each sensor and is used to restore the default configuration.

Definition

```
#define RKADK_DEFPARAM_PATH_SENSOR_PREFIX "/oem/usr/etc/rkadk_defsetting_sensor"
```

Notice

- The sensor configuration file prefix defaults to rkadk_defsetting_sensor_n.ini, _n is the sensor camera ID, and the serial number starts from 0.

12.3.3 RKADK_PARAM_PATH

Description

Global ini configuration file path, which stores common configurations of all sensors.

Definition

```
#define RKADK_PARAM_PATH "/data/rkadk/rkadk_setting.ini"
```

12.3.4 RKADK_PARAM_PATH_SENSOR_PREFIX

Description

Sensor ini configuration file path, which stores the unique configuration of each sensor. Used to save new configurations when switching recording resolution, Codec type and other operations.

Definition

```
#define RKADK_PARAM_PATH_SENSOR_PREFIX "/data/rkadk/rkadk_setting_sensor"
```

Notice

- The sensor configuration file prefix defaults to rkadk_setting_sensor_n.ini, _n is the sensor camera ID and the serial number starts from 0.

12.3.5 RKADK_PARAM_TYPE_E

Description

Define parameter type enumeration type.

Definition

```
typedef enum {
  /* Cam Dependent Param */
  RKADK_PARAM_TYPE_FPS,             /* framerate */
  RKADK_PARAM_TYPE_GOP,             /* gop */
  RKADK_PARAM_TYPE_RES,             /* specify RKADK_PARAM_RES_E(record) */
  RKADK_PARAM_TYPE_PHOTO_RES,       /* specify RKADK_PARAM_RES_E(photo) */
  RKADK_PARAM_TYPE_STREAM_RES,      /* specify RKADK_PARAM_RES_CFG_S(record_main,
record_sub, photo, stream, live) */
  RKADK_PARAM_TYPE_CODEC_TYPE,      /* specify RKADK_PARAM_CODEC_CFG_S(record) */
  RKADK_PARAM_TYPE_BITRATE,         /* encode bitrate,specify
RKADK_PARAM_BITRATE_S */
  RKADK_PARAM_TYPE_FLIP,            /* bool */
  RKADK_PARAM_TYPE_MIRROR,          /* bool */
  RKADK_PARAM_TYPE_LDC,             /* ldc level [0,255] */
  RKADK_PARAM_TYPE_ANTIFOG,         /* antifog value, [0,10] */
  RKADK_PARAM_TYPE_WDR,             /* wdr level, [0,10] */
  RKADK_PARAM_TYPE_HDR,             /* 0: normal, 1: HDR2, 2: HDR3, [0,2] */
  RKADK_PARAM_TYPE_RECORD_TYPE,     /* specify RKADK_REC_TYPE_E */
  RKADK_PARAM_TYPE_RECORD_TIME,     /* specify RKADK_PARAM_REC_TIME_S, record
time(s) */
  RKADK_PARAM_TYPE_PRE_RECORD_TIME, /* pre record time, unit in second(s) */
  RKADK_PARAM_TYPE_PRE_RECORD_MODE, /* pre record mode, specify
MUXER_PRE_RECORD_MODE_E */
```

```
  RKADK_PARAM_TYPE_SPLITTIME,       /* specify RKADK_PARAM_REC_TIME_S, manual
splite time(s) */
  RKADK_PARAM_TYPE_FILE_CNT,        /* record file count, maximum
RECORD_FILE_NUM_MAX */
  RKADK_PARAM_TYPE_LAPSE_INTERVAL,  /* specify RKADK_PARAM_REC_TIME_S, lapse
interval(s) */
  RKADK_PARAM_TYPE_LAPSE_MULTIPLE,  /* lapse multiple */
  RKADK_PARAM_TYPE_JPEG_SLICE,      /* enable/disable JPEG slice */
  RKADK_PARAM_TYPE_SLICE_HEIGHT,    /* set JPEG slice height */
  RKADK_PARAM_TYPE_INPUT_FMT,       /* set vi input format */
  RKADK_PARAM_TYPE_INPUT_BUFCNT,    /* set vi input buffer count */
  RKADK_PARAM_TYPE_ENABLE_AUDIO,    /* enable record audio */
  RKADK_PARAM_TYPE_MUTE,            /* mute specifies sensor, bool */

  /* COMM Dependent Param */
  RKADK_PARAM_TYPE_REC_MUTE,        /* record audio mute, bool */
  RKADK_PARAM_TYPE_VOLUME,          /* speaker volume, [0,100] */
  RKADK_PARAM_TYPE_MIC_VOLUME,      /* mic volume, [0,100] */
  RKADK_PARAM_TYPE_BUTT
} RKADK_PARAM_TYPE_E;
```

Members

Member Name	Description
RKADK_PARAM_TYPE_FPS	Frame rate
RKADK_PARAM_TYPE_GOP	I frame interval, RKADK_PARAM_GOP_S
RKADK_PARAM_TYPE_RES	Video resolution, RKADK_PARAM_RES_E
RKADK_PARAM_TYPE_PHOTO_RES	Photo resolution, RKADK_PARAM_RES_E
RKADK_PARAM_TYPE_STREAM_RES	Sets the resolution of the stream enumeration specified by RKADK_STREAM_TYPE_E
RKADK_PARAM_TYPE_CODEC_TYPE	Video encoding type, RKADK_PARAM_CODEC_CFG_S
RKADK_PARAM_TYPE_BITRATE	Bitrate, RKADK_PARAM_BITRATE_S
RKADK_PARAM_TYPE_FLIP	Flip up and down
RKADK_PARAM_TYPE_MIRROR	Mirror left and right
RKADK_PARAM_TYPE_LDC	Distortion correction[0,255]
RKADK_PARAM_TYPE_ANTIFOG	Dehaze[0,10]
RKADK_PARAM_TYPE_WDR	Wide dynamic range[0,10]
RKADK_PARAM_TYPE_HDR	High dynamic range imaging[0,10]
RKADK_PARAM_TYPE_RECORD_TYPE	Recording type, RKADK_REC_TYPE_E
RKADK_PARAM_TYPE_RECORD_TIME	Recording duration, recording main stream and sub-stream support setting different durations, RKADK_PARAM_REC_TIME_S
RKADK_PARAM_TYPE_PRE_RECORD_TIME	Pre-recording duration
RKADK_PARAM_TYPE_PRE_RECORD_MODE	Pre-recording mode, 0: no pre-recording, 1: manually split the pre-recording, 2: first file pre-recording, 3: all files pre-recording
RKADK_PARAM_TYPE_SPLITTIME	Manually split the recording duration. The main video stream and sub-stream of the video support setting different durations, RKADK_PARAM_REC_TIME_S
RKADK_PARAM_TYPE_FILE_CNT	Number of simultaneous recording files, the maximum is 2
RKADK_PARAM_TYPE_LAPSE_INTERVAL	Time-lapse recording duration, recording main stream and sub-stream support setting different durations, RKADK_PARAM_REC_TIME_S

Member Name	Description
RKADK_PARAM_TYPE_LAPSE_MULTIPLE	The multiple relationship between the playback duration of the time-lapse video file and the actual screen content time
RKADK_PARAM_TYPE_JPEG_SLICE	Whether to enable JPEG Slice
RKADK_PARAM_TYPE_SLICE_HEIGHT	JPEG Slice high
RKADK_PARAM_TYPE_INPUT_FMT	VI input format
RKADK_PARAM_TYPE_INPUT_BUFCNT	VI buffer count
RKADK_PARAM_TYPE_ENABLE_AUDIO	Whether to enable specifies sensor's recording audio channel
RKADK_PARAM_TYPE_MUTE	Whether to enable specifies sensor's recording mute
RKADK_PARAM_TYPE_REC_MUTE	Whether to enable all recording mute
RKADK_PARAM_TYPE_VOLUME	Speaker volume[0,100]
RKADK_PARAM_TYPE_MIC_VOLUME	Microphone volume[0,100]

Notice

- Antifog, WDR, HDR and other camere hardware related settings, in addition to calling RKADK_PARAM_SetCamParam to set the ini, you also need to call the ISP corresponding interface to make it actually take effect.
- RKADK_PARAM_TYPE_LAPSE_MULTIPLE: the relationship between the playback duration of the time-lapse video file and the actual screen content time is related to the frame rate. For example, the frame rate of ordinary video is 30fps, and the time-lapse video is 1fps, then the multiple is 30.
- When RV1126/RV1109 switches resolutions and the photo resolution is not set to the maximum supported resolution of the sensor, it needs to be consistent with the Record main stream resolution.
- RKADK_PARAM_TYPE_REC_MUTE: Should be used together with RKADK_RECORD_GetAencChn and RK_MPI_AENC_SetMute.

Related data types and interfaces

RKADK_PARAM_GetCamParam

RKADK_PARAM_SetCamParam

RKADK_PARAM_GetCommParam

RKADK_PARAM_SetCommParam

12.3.6 RKADK_PARAM_RES_E

Description

Defines the playback event enumeration type.

Definition

```
typedef enum {
  RKADK_RES_480P = 0, /* 848*480 */
  RKADK_RES_720P,     /* 1280*720 */
  RKADK_RES_1080P,    /* 1920*1080 */
  RKADK_RES_1296P,    /* 2304*1296 */
  RKADK_RES_1440P,    /* 2560*1440 */
  RKADK_RES_1520P,    /* 2688*1520 */
  RKADK_RES_1600P,    /* 2560*1600 */
  RKADK_RES_1620P,    /* 2880*1616, height 8 alignment */
  RKADK_RES_1944P,    /* 2592*1944 */
  RKADK_RES_2160P,    /* 3840*2160 */
  RKADK_RES_3456P,    /* 4608*3456 */
  RKADK_RES_4000P,    /* 6000*4000 */
  RKADK_RES_4480P,    /* 6720*4480 */
  RKADK_RES_4275P,    /* 7600*4275 */
  RKADK_RES_4320P,    /* 7680*4320 */
  RKADK_RES_3672P,    /* 6528*3672 */
  RKADK_RES_BUTT,
} RKADK_PARAM_RES_E;
```

Related data types and interfaces

RKADK_PARAM_GetResolution

RKADK_PARAM_GetResType

12.3.7 RKADK_STREAM_TYPE_E

Description

Define the stream enumeration type.

Definition

```
typedef enum {
  RKADK_STREAM_TYPE_SENSOR,
  RKADK_STREAM_TYPE_VIDEO_MAIN,
  RKADK_STREAM_TYPE_VIDEO_SUB,
  RKADK_STREAM_TYPE_SNAP,
  RKADK_STREAM_TYPE_PREVIEW,
  RKADK_STREAM_TYPE_LIVE,
  RKADK_STREAM_TYPE_DISP,
  RKADK_STREAM_TYPE_BUTT
} RKADK_STREAM_TYPE_E;
```

Members

Member Name	Description
RKADK_STREAM_TYPE_VIDEO_MAIN	Video main stream
RKADK_STREAM_TYPE_VIDEO_SUB	Video sub-stream
RKADK_STREAM_TYPE_SNAP	Take photos
RKADK_STREAM_TYPE_PREVIEW	Remote preview
RKADK_STREAM_TYPE_LIVE	Live broadcast
RKADK_STREAM_TYPE_DISP	Local Preview
RKADK_STREAM_TYPE_SENSOR	Sensor

Related data types and interfaces

RKADK_PARAM_GetVencChnId

12.3.8 RKADK_PARAM_CODEC_CFG_S

Description

Define the encoding type configuration structure.

Definition

```
typedef struct {
  RKADK_STREAM_TYPE_E enStreamType;
  RKADK_CODEC_TYPE_E enCodecType;
} RKADK_PARAM_CODEC_CFG_S;
```

Members

Member Name	Description
enStreamType	stream type
enCodecType	encoding type

Related data types and interfaces

RKADK_CODEC_TYPE_E

RKADK_STREAM_TYPE_E

12.3.9 RKADK_PARAM_BITRATE_S

Description

Define the bitrate configuration structure.

Definition

```
typedef struct {
  RKADK_STREAM_TYPE_E enStreamType;
  RKADK_U32 u32Bitrate;
} RKADK_PARAM_BITRATE_S;
```

Members

Member Name	Description
enStreamType	stream type
u32Bitrate	Bitrate

Related data types and interfaces

RKADK_STREAM_TYPE_E

12.3.10 RKADK_PARAM_REC_TIME_S

Description

Define the recording duration configuration structure.

Definition

```
typedef struct {
  RKADK_STREAM_TYPE_E enStreamType;
  RKADK_U32 time;
} RKADK_PARAM_REC_TIME_S;
```

Members

Member Name	Description
enStreamType	Stream type
time	Recording duration

Related data types and interfaces

RKADK_STREAM_TYPE_E

12.3.11 RKADK_PARAM_GOP_S

Description

Define the VENC I frame interval configuration structure.

Definition

```
typedef struct {
  RKADK_STREAM_TYPE_E enStreamType;
  RKADK_U32 u32Gop;
} RKADK_PARAM_GOP_S;
```

Members

Member Name	Description
enStreamType	stream type
u32Gop	I frame interval

Related data types and interfaces

RKADK_STREAM_TYPE_E

12.3.12 RKADK_VQE_MODE_E

Description

Defines the audio input sound quality enhancement enumeration type

Definition

```
typedef enum {
  RKADK_VQE_MODE_AI_TALK = 0,
  RKADK_VQE_MODE_AI_RECORD,
  RKADK_VQE_MODE_BUTT
} RKADK_VQE_MODE_E;
```

Members

Member Name	Description
RKADK_VQE_MODE_AI_TALK	Enable AEC, ANR, AGC
RKADK_VQE_MODE_AI_RECORD	Enable ANR

12.3.13 RKADK_MUXER_FILE_TYPE_E

Description

Define recording file type enumeration

Definition

```
typedef enum rkMUXER_TYPE_E {
  RKADK_MUXER_TYPE_MP4 = 0,
  RKADK_MUXER_TYPE_MPEGTS,
  RKADK_MUXER_TYPE_FLV,
  RKADK_MUXER_TYPE_BUTT
} RKADK_MUXER_FILE_TYPE_E;
```

Members

Member Name	Description
RKADK_MUXER_TYPE_MP4	MP4
RKADK_MUXER_TYPE_MPEGTS	Reserved
RKADK_MUXER_TYPE_FLV	FLV

12.3.14 RKADK_MUXER_PRE_RECORD_MODE_E

Description

Define prerecorded mode enum

Definition

```
typedef enum {
  RKADK_MUXER_PRE_RECORD_NONE = 0,
  RKADK_MUXER_PRE_RECORD_MANUAL_SPLIT, /* manual split file prerecord */
  RKADK_MUXER_PRE_RECORD_SINGLE        /* first file prerecord */
} RKADK_MUXER_PRE_RECORD_MODE_E;
```

Members

Member Name	Description
RKADK_MUXER_PRE_RECORD_NONE	No pre-recording
RKADK_MUXER_PRE_RECORD_MANUAL_SPLIT	Manually split file pre-recording
RKADK_MUXER_PRE_RECORD_SINGLE	First file pre-recorded

12.3.15 RKADK_MIC_TYPE_E

Description

Define the audio device channel mode type

Definition

```
typedef enum {
  RKADK_MIC_TYPE_LEFT = 0,
  RKADK_MIC_TYPE_RIGHT,
  RKADK_MIC_TYPE_BOTH,
  RKADK_MIC_TYPE_BUTT
} RKADK_MIC_TYPE_E;
```

Members

Member Name	Description
RKADK_MIC_TYPE_LEFT	Left channel sound
RKADK_MIC_TYPE_RIGHT	Right channel sound
RKADK_MIC_TYPE_BOTH	Dual-channel

12.4 INI File Introduction

12.4.1 Global INI Configuration File

```
[version]
version                         = 2.2.0      /* Version */

/* common parameters */
[common]
sensor_count                    = 2          /* Number of Sensors*/
rec_mute                        = FALSE      /* Whether to enable video mute */
speaker_volume                  = 80         /* Speaker volume,[0,100] */
mic_volume                      = 80         /* Microphone volume,[0,100] */
vpss_devcie                     = 1          /* VPSS hardware device type, 0:GPU,
1:RGA */

/* Audio parameters */
[audio]
ai_audio_node                   = hw:0,0     /* AI device node */
ao_audio_node                   = hw:0,0     /* AO device node */
ai_depth                        = 1          /* AI depth */
bit_width                       = 1          /* Sampling accuracy */
channels                        = 1          /* Channels */
mic_type                        = 0          /* Audio device channel mode,
specifically RKADK_MIC_TYPE_E, 0: left channel, 1: right channel, 2: dual
channel*/
samplerate                      = 16000      /* Sample rate   */
samples_per_frame               = 576        /* Number of samples per frame*/
bitrate                         = 64000      /* Bitrate */
vqe_mode                        = 1          /* Configure audio input sound
quality enhancement, specifically RKADK_VQE_MODE_E*/
vqe_config_path                 = /oem/usr/share/vqefiles/config_aivqe.json /*vqe
configuration file path */
```

```
codec_type                      = 8          /* Record and Live Audio encoding
types, adapted to MP3 by default, specifically RKADK_CODEC_TYPE_E */
```

12.4.2 Sensor INI Configuration Files

```
[sensor]
used_isp                        = TRUE       /* Has the Sensor used ISP? */
max_width                       = 2688       /* Max width */
max_height                      = 1520       /* Max height    */
framerate                       = 30         /* Framerate*/
flip                            = FALSE      /* Flip up and down*/
mirror                          = FALSE      /* Mirror left and right */
ldc                             = 0          /* Lens distortion correction,
[0,255] */
wdr                             = 0          /* Wide dynamic range, [0,10] */
hdr                             = 0          /* High dynamic range imaging, [0,10]
*/
antifog                         = 0          /* Remove fog, [0,10] */
enable_wrap                     = FALSE      /* Whether the VI enables wrap */
wrap_buf_line                   = 1620       /* The line height of the wrap buffer
*/

/* VI channel configuration parameters */
[vi.0]
chn_id                          = 0                  /* Channel ID */
device_name                     = rkispp_m_bypass    /* Video node path*/
buf_cnt                         = 4                  /* The total number of buffer
blocks in the output channel */
depth                           = 0                  /* VI depth depth  */
width                           = 2688               /* Video width   */
height                          = 1520               /* Video height  */
pix_fmt                         = FBC0               /* VI output format*/
module                          = RECORD_MAIN|PHOTO  /* The usage module of this
VI,Options: NONE/RECORD_MAIN/RECORD_SUB/PREVIEW/PHOTO/LIVE/DISP */

[vi.1]
chn_id                          = 1
device_name                     = rkispp_scale0
buf_cnt                         = 4
depth                           = 0
width                           = 0
height                          = 0
pix_fmt                         = NV12
module                          = RECORD_MAIN|PHOTO

[vi.2]
chn_id                          = 2
device_name                     = rkispp_scale1
buf_cnt                         = 2
depth                           = 0
width                           = 0
height                          = 0
pix_fmt                         = NV12
```

```
module                          = NONE

[vi.3]
chn_id                          = 3
device_name                     = rkispp_scale2
buf_cnt                         = 4
depth                           = 0
width                           = 848
height                          = 480
pix_fmt                         = NV12
module                          = RECORD_SUB|PREVIEW|LIVE|DISP

/* Record parameters */
[record]
record_type                     = 0          /* Recording type, specifically
RKADK_REC_TYPE_E*/
file_type                       = 0          /* Video file type, specifically
RKADK_MUXER_FILE_TYPE_E */
pre_record_time                 = 0          /* Pre-recording duration */
pre_record_mode                 = 0          /* Pre-recorded mode */
lapse_multiple                  = 30         /* The multiple relationship between
the playback duration of a time-lapse video file and the actual screen content
duration */
file_num                        = 1          /* Number of files recorded
simultaneously, maximum 2 */
switch_res                      = TRUE       /* Whether to switch resolution */
enable_audio                    = TRUE       /* Whether to enable recording */

/* Record 0 VENC parameters of Main stream  */
[record.0]
record_time                     = 60         /* Recording duration*/
splite_time                     = 60         /* Manually divide video duration */
lapse_interval                  = 60         /* Time-lapse recording duration */
width                           = 2688       /* Video width */
height                          = 1520       /* Video height */
bufsize                         = 10379776   /* The buffer size of the code stream
*/
framerate                       = 30         /* Venc framerate */
bitrate                         = 8294400    /* Bitrate */
gop                             = 30         /* I frame interval */
profile                         = 100        /* Encoder profile */
codec_type                      = 0          /* Encoding type,
specificallyRKADK_CODEC_TYPE_E */
venc_chn                        = 0          /* VENC channel ID */
vpss_grp                        = 0          /* VPSS GROUP ID*/
vpss_chn                        = 0          /* VPSS channel ID */
post_aiisp                      = FALSE      /* Whether to enbale Post AI ISP */
rc_mode                         = CBR        /* Encoding protocol type, supports
CBR, VBR, AVBR*/
max_qp                          = -1         /* QP maximum value, value range [1,
51], -1 is the default value */
min_qp                          = -1         /* QP minimum value, value range [1,
max_qp], -1 is the default value */
i_min_qp                        = -1
i_frame_min_qp                  = -1
full_range                      = TRUE
```

```
scaling_list                    = FALSE
hier_qp_en                      = FALSE
hier_qp_delta                   = -3,0,0,0
hier_frame_num                  = 3,0,0,0

/* VENC parameter of the substream Record 1, when file_num = 1, rec.1 does not
need to be configured*/
[record.1]
record_time                     = 60
splite_time                     = 60
lapse_interval                  = 60
width                           = 848
height                          = 480
bufsize                         = 2367488
bitrate                         = 407040
framerate                       = 30
gop                             = 30
profile                         = 100
codec_type                      = 0
venc_chn                        = 1
vpss_grp                        = 0          /* VPSS GROUP ID*/
vpss_chn                        = 0          /* VPSS channel ID */
post_aiisp                      = FALSE      /* Whether to enbale Post AI ISP */
rc_mode                         = VBR
max_qp                          = 48
min_qp                          = 8
i_min_qp                        = -1
i_frame_min_qp                  = -1
full_range                      = TRUE
scaling_list                    = TRUE
hier_qp_en                      = TRUE
hier_qp_delta                   = -3,0,0,0
hier_frame_num                  = 3,0,0,0

/* Photo VENC parameters */
[photo]
image_width                     = 3840       /* Image width */
image_height                    = 2160       /* Image_height  */
venc_chn                        = 2          /* VENC channel ID */
vpss_grp                        = 0          /* VPSS GROUP ID*/
vpss_chn                        = 0          /* VPSS channel ID */
post_aiisp                      = FALSE      /* Whether to enbale Post AI ISP */
enable_combo                    = FALSE      /* Enable the Combo attribute of the
encoding channel */
combo_venc_chn                  = 0          /* Combo data source channel */
qfactor                         = 50         /* For the detailed meaning, please
refer to the RFC2435 protocol. The system default value is 70 and the value range
is [1, 99] */
switch_res                      = TRUE       /* Whether to switch resolution */
jpeg_slice                      = FALSE      /* Whether to enable JPEG Slice */
slice_height                    = 0          /* The height of JPEG Slice must not
be greater than max_slice_height*/
max_slice_width                 = 0          /* Maximum width of JPEG Slice */
max_slice_height                = 0          /* Maximum height of JPEG Slice*/

/* Remote preview of VENC parameters */
```

```
[preview]
width                           = 848        /* Video width */
height                          = 480        /* Video height */
bufsize                         = 2367488
bitrate                         = 407040     /* Bitrate */
framerate                       = 30         /* Venc framerate  */
gop                             = 30         /* I frame interval */
profile                         = 100        /* Encoder profile */
codec_type                      = 0          /* Coding type, specifically
RKADK_CODEC_TYPE_E */
venc_chn                        = 1          /* Venc channel ID */
vpss_grp                        = 0          /* VPSS GROUP ID */
vpss_chn                        = 0          /* VPSS  channel ID */
post_aiisp                      = FALSE      /* Whether to enbale Post AI ISP */
rc_mode                         = VBR        /* Encoding protocol type, supports
CBR, VBR, AVBR */
max_qp                          = 48         /* QP maximum value, value range [1,
51] */
min_qp                          = 8          /* QP minimum value, value range [1,
min_qp] */
i_min_qp                        = -1
i_frame_min_qp                  = -1
full_range                      = TRUE
scaling_list                    = TRUE
hier_qp_en                      = TRUE
hier_qp_delta                   = -3,0,0,0
hier_frame_num                  = 3,0,0,0

/* Live VENC parameters */
[live]
width                           = 1280
height                          = 720
bufsize                         = 2367488
bitrate                         = 4194304
framerate                       = 30
gop                             = 30
profile                         = 100
codec_type                      = 0
venc_chn                        = 1
vpss_grp                        = 0
vpss_chn                        = 0
post_aiisp                      = FALSE
rc_mode                         = VBR
max_qp                          = 48
min_qp                          = 8
i_min_qp                        = -1
i_frame_min_qp                  = -1
full_range                      = TRUE
scaling_list                    = TRUE
hier_qp_en                      = TRUE
hier_qp_delta                   = -3,0,0,0
hier_frame_num                  = 3,0,0,0

/* Local preview parameters */
[display]
x                               = 0              /* Display X coordinate */
```

```
y                               = 0                 /* Display y coordinate */
width                           = 720               /* Display the width  */
height                          = 1280              /* Display the height */
rotaion                         = 90                /* Degrees of rotation,
Options: 0:0, 1:90, 2:180, 3:270 */
vpss_grp                        = 0                 /* VPSS GROUP ID */
vpss_chn                        = 0                 /* VPSS channel ID */
img_type                        = RGB888            /* Video output format */
vo_device                       = 0                 /* Display output device number
*/
vo_layer                        = 0                 /* Video output video layer
number */
vo_chn                          = 0                 /* VO channel ID */
frame_rate                      = 30                /* VO frame rate */
intf_type                       = default           /* Show interface type,
Options:  MIPI, HDMI, EDP, VGA, DP, HDMI_EDP, LCD, default*/
splice_mode                     = RGA               /* Splice mode, Options: RGA,
GPU, BYPASS */
```

12.4.3 INI Configuration Precaution

- Configure the default INI file path through environment variables, for example: export rkadk_default_ini_path=/oem/usr/etc.
- Set the INI file path through RKADK_PARAM_Init API.
- The versions in rkadk_defsetting.ini and rkadk_setting.ini must be consistent, otherwise version detection will fail and the default INI configuration will be used. When the SDK is updated, the ini parameters may increase or decrease, so you need to pay attention at this time.
- The sensor_count represents the actual number of sensors used. It is set according to actual use and cannot be greater than RKADK_MAX_SENSOR_CNT. Currently, 3 sensors have been actually debugged.
- If the sensor uses ISP, `used_isp` must be configured to TRUE, and scaling is achieved directly by configuring the ISP node resolution. There is no need to configure the VPSS channel. vpss_grp and vpss_chn are both configured to 0; if the sensor does not use ISP, `used_isp` must be configured to FALSE, and configured the corresponding vpss_grp and vpss_chn, VPSS is used internally for scaling.
- When the resolution of recording, remote preview, live broadcast, etc. are the same, it is recommended to reuse VENC (VENC parameters are configured to be the same) to improve bandwidth and memory utilization.
- When VENC is reused or the VENC resolution is the same, it is recommended to reuse VPSS channels to improve bandwidth utilization.
- The gop is recommended to be configured the same as framerate to ensure that there is one I frame every second.

13. Examples

Function examples are provided below, and precautions for use are as follows:

- Before running the examples, make sure that no other application occupies the node used by the example, such as mediaserver and ispserver.
- The default parameters of the examples are adapted to RK EVBs. When the hardware are different from our EVB, customers need to specify command parameters themselves or adjust the code.

13.1 rkadk_record_test

Description

Record test.

Code path

rkadk/examples/rkadk_record_test.c

Quick use

```
./rkadk_record_test
```

Option

Options	Description	Default
-a	Enable the built-in ISP function. Enter this option to enable the built-in ISP function. If there are no parameters, the default value will be used. The parameter is the path to the folder where the aiq file is located.	/etc/iqfiles
-I	Camera ID	0
-m	Enable dual sensor test, options: 0(isp sensor), 1(all isp sensors), 2(isp+ahd sensors)	0
-p	The ini configuration file directory path	/data/rkadk
-k	Whether the video file is sliced into I frames	Not sliced

13.2 rkadk_photo_test

Description

Photo testing.

Code path

/rkadk/examples/rkadk_photo_test.c

Quick use

```
./rkadk_photo_test
```

Option

Options	Description	Default
-a	Enable the built-in ISP function. Enter this option to enable the built-in ISP function. If there are no parameters, the default value will be used. The parameter is the path to the folder where the aiq file is located.	/etc/iqfiles
-I	Camera ID	0
-p	The ini configuration file directory path	/data/rkadk
-t	Data type of the obtained JPG image	NV12
-m	Enable dual sensor test, options: 0(isp sensor), 1(all isp sensors), 2(isp+ahd sensors)	0
-o	The osd file path	NULL
-W	The osd width	0
-H	The osd high	0

13.3 rkadk_stream_test

Description

Get the audio stream, encode it, and output it to a file; get the video stream, encode it, and output it to a file.

Code path

/rkadk/examples/rkadk_stream_test.c

Quick use

```
./rkadk_stream_test
```

Option

Options	Description	Default
-a	Enable the built-in ISP function. Enter this option to enable the built-in ISP function. If there are no parameters, the default value will be used. The parameter is the path to the folder where the aiq file is located.	/etc/iqfiles
-I	Camera ID	0
-M	Test mode: audio, video	audio
-e	Encoding type	pcm
-o	Output file path	/tmp/ai.pcm
-p	The ini configuration file directory path	/data/rkadk
-m	Enable dual Sensor test, options: 0(isp sensor), 1(all isp sensors), 2(isp+ahd sensors)	0

13.4 rkadk_player_test

Description

Local file playback test.

Code path

/rkadk/examples/rkadk_player_test.c

Quick use

```
./rkadk_player_test
```

Option

Options	Description	Default
-i	Playback file path	/etc/bsa_file/8k8bpsMono.wav
-x	Video display starting x coordinate	0
-y	Video display starting y coordinate	0
-W	Video display width	Screen physical width
-H	Video display height	Screen physical height
-r	Video rotation angle, option: 0, 90, 180, 270	0
-p	The ini configuration file directory path	/data/rkadk
-m	Video mirror	disable
-f	Video flip	disable
-a	Whether to enable audio playback, option: 0 (disable), 1 (enable)	1
-v	Whether to enable Video playback	disable
-s	Set layer synthesis method, option: 0(RGA), 1(GPU), 2(ByPass)	0
-P	Screen display pixel format, option: 0(RGB888), 1(NV12), 2(RGB565)	0
-I	Display interface type, option: 0(DEFAILT), 1(MIPI), 2(LCD)	1106: 0, other platforms: 1
-F	Refresh frame rate	30
-t	RTSP transmission protocol, option: 0(udp), 1(tcp)	0
-b	Enable black screen after playing	disable
-T	RTSP socket I/O operation timeout exit time, unit ms	Blocking, no timeout
-l	VO layer ID	0
-O	Vdec output buffer number	3 [1, 8]
-D	Enable third-party demuxer library	disable

Notice

- When playing video files, -v is required to enable Video playback.
- Mirror/flip cannot be set at the same time as rotation.

13.5 rkadk_thumb_test

Get file thumbnail test.

Code path

/rkadk/examples/rkadk_thumb_test.c

Quick use

```
Get MP4 thumbnail: ./rkadk_thumb_test -i test_file.mp4
Get JPG thumbnail: ./rkadk_thumb_test -i test_file.jpg -f jpg -t MPF1
```

Option

Options	Description	Default
-i	Test file path	None
-f	File format: mp4, jpg	mp4
-t	JPG thumbnail types: DCF, MPF1, MPF2	DCF
-T	Output thumbnail type: JPG, NV12, RGB565, RGB888	JPG
-W	Thumbnail width	Get from ini
-H	Thumbnail height	Get from ini

13.6 rkadk_rtsp_test

RTSP live test.

Code path

/rkadk/examples/rkadk_rtsp_test.c

Quick use

```
1. Start wifi on the board end
2. Run the board end./rkadk_rtsp_test
3. Open VLC on PC -> Media -> Open network streaming -> Enter the network URL:
rtsp://board end ip address/live/main_stream
```

Option

Options	Description	Default
-a	Enable the built-in ISP function. Enter this option to enable the built-in ISP function. If there are no parameters, the default value will be used. The parameter is the path to the folder where the aiq file is located.	/etc/iqfiles
-I	Camera ID	0
-p	The ini configuration file directory path	/data/rkadk
-o	The osd file path	/userdata/rkadk_ARGB8888

13.7 rkadk_rtmp_test

RTMP live test.

Code path

/rkadk/examples/rkadk_rtmp_test.c

Quick use

```
1. Start wifi on the board
2. Run ./rkadk_rtmp_test on the board
3. Open VLC on the PC -> Media -> Open network streaming -> Enter the network
URL: rtmp://The ip address of the board::1935/live/substream
```

Option

Options	Description	Default
-a	Enable the built-in ISP function. Enter this option to enable the built-in ISP function. If there are no parameters, the default value will be used. The parameter is the path to the folder where the aiq file is located.	/etc/iqfiles
-I	Camera ID	0
-p	The ini configuration file directory path	/data/rkadk

13.8 rkadk_storage_test

Description

Storage module testing.

Code path

/rkadk/examples/rkadk_storage_test.c

Quick use

```
./rkadk_storage_test
```

Option

None

Notice

- This test generates an mp4 file with all 0s written in it to test functions and interfaces such as automatic deletion and file list obtaining. The mp4 file has no actual data and cannot be played.

13.9 rkadk_disp_test

Local preview testing.

Code path

/rkadk/examples/rkadk_disp_test.c

Quick use

```
./rkadk_disp_test
```

Option

Options	Description	Default
-a	Enable the built-in ISP function. Enter this option to enable the built-in ISP function. If there are no parameters, the default value will be used. The parameter is the path to the folder where the aiq file is located.	/etc/iqfiles
-I	Camera ID	0
-p	The ini configuration file directory path	/data/rkadk

13.10 rkadk_ui_test

UI overlay testing.

Code path

/rkadk/examples/rkadk_ui_test.c

Quick use

```
./rkadk_ui_test
```

Option

Options	Description	Default
-a	Enable the built-in ISP function. Enter this option to enable the built-in ISP function. If there are no parameters, the default value will be used. The parameter is the path to the folder where the aiq file is located.	/etc/iqfiles
-I	Camera ID	0
-p	The ini configuration file directory path	/data/rkadk
-W	Display area width	720
-H	Display area height	1280
-f	Screen display pixel format, option: 0(RGB888), 1(NV12), 2(RGB565), 3(RGB444)	0